Chapter 1. Statistical Learning, Data and Exploration

Lectures 1–3 (overview of statistics/ML/AI, statistical learning, data as probabilistic evidence, assumptions, data structures, EDA)

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1 Statistics, Machine Learning, AI and Data Mining

1.1 Where are we? A statistician's view

- **Statistics**: modelling and analysing the data-generating process under uncertainty.
- **Statistical learning**: estimating $P(Y \mid X)$ from limited samples.
- **Machine learning**: optimisation for prediction.
- **Data mining**: discovery of patterns and structure in complex data.

AI is the broadest field (intelligent systems and decision making); ML is the subset that learns from data; deep learning is one kind of ML; statistics underpins many ML methods and models uncertainty; data science uses statistics and ML to extract insight from data.

Example: one problem, four perspectives — customer churn

AI: build an intelligent system that decides retention strategies. **ML**: learn $\hat{f}(X)$ to predict the churn probability. **Statistics**: which variables matter, and how certain are we? **Data mining**: discover hidden segments and behavioural structure. Same data, same problem, different questions.

Data are not just numbers but *traces of a hidden generative process*. Intelligence is not accuracy alone and learning is not memorisation: it is an attempt to understand that process.

2 What Is Statistical Learning?

Statistical learning studies how to estimate unknown probabilistic relationships from finite data.

- **Supervised learning**: estimate $P(Y \mid X)$ from labelled data.
- **Unsupervised learning**: uncover structure in $P(X)$ without labels.

The course focuses less on algorithms as recipes and more on the principles behind them.

Definition: notation

Outcome (response, target) Y; predictor vector $X = (X_1, \ldots, X_p)$; observed data $\{(X_i, Y_i)\}_{i=1}^n$. The problem is **regression** if Y is quantitative and **classification** if Y is categorical.

Supervised learning: model $P(Y \mid X)$	Unsupervised learning: model $P(X)$
prediction: perform well on unseen data **inference**: understand how X influences Y **uncertainty**: reliability of predictions and estimates	**structure estimation**: properties of $P(X)$ **clustering**: latent groups, multimodality **representation learning**: low-dimensional summaries of variability

Learning is not just fitting — it is *generalisation under uncertainty*.

2.1 Applications

The same framework (estimate an unknown relationship from data) covers medical risk prediction (probability of a heart attack), text classification (spam detection), image recognition (handwritten digits) and economic modelling (income versus demographic variables).

Example: spam data

4601 e-mails sent to one person ("George" at HP Labs, before 2000), each labelled spam or e-mail. Features: relative frequencies of the 57 most common words and punctuation marks. Words with the largest spam/e-mail difference: `george` (0.00 vs 1.27%), `you` (2.26 vs 1.27), `hp` (0.02 vs 0.90), `free` (0.52 vs 0.07), `!` (0.51 vs 0.11), `edu` (0.01 vs 0.29), `remove` (0.28 vs 0.01). Goal: a customised spam filter.

3 Data as Probabilistic Evidence

3.1 Sample versus population

Data are observations, not the population; observations are realisations of random variables, which arise from an unknown data-generating process. Data are **stochastic evidence, not absolute truth**.

Definition: the inferential setting

Observed sample $\mathcal{D}_n = \{Z_1, \ldots, Z_n\}$, population distribution $Z \sim P_0$, target quantity $\theta(P_0)$ (a centre, a dispersion, a regression coefficient, ...). The sample is finite; the population is conceptual.

Sampling variability. Estimators are random because samples are random: tossing the same coin n times and repeating the experiment gives a different $\hat{p} = \bar{Z}$ each time, with $\mathrm{Var}(\hat{p}) = p(1-p)/n$.

3.2 Assumptions and identifiability

A finite data set is consistent with many possible underlying mechanisms; data alone do not uniquely determine the target, $\mathcal{D}_n \not\Rightarrow \theta(P_0)$. To make learning possible we restrict the set of possible worlds, $P_0 \in \mathcal{P}_{\text{model}}$; within this restricted class identification and valid procedures become possible.

Example: five rainy days

The last five days were Rain, Rain, Rain, Rain, Rain. Does it always rain? Did the rainy season just start? Were we unlucky this week? The same data admit several explanations; only an assumption (e.g. i.i.d. days, or a seasonal model) picks one.

Definition: data-generating process

$$Y = f(X) + \varepsilon,$$

where f is the underlying mechanism and ε captures randomness and unobserved factors.

Statistical modelling specifies a **class** of functions for f; different classes encode different structural assumptions and determine model flexibility, and they require different mathematical tools:

Function class	Example	Tools
linear models	$f(x) = x^\top \beta$	linear algebra, projections, least squares
parametric models	$f(x) = f(x;\theta)$, $\theta \in \Theta$	likelihood theory, asymptotics, Fisher information
RKHS (kernel) models	$f \in \mathcal{H}_K$	Hilbert spaces, kernels, representer theorem
nonparametric classes	Sobolev, Hölder balls	functional analysis, approximation theory

4 Data Structures and Their Implications

4.1 The i.i.d. assumption

Classical setting: $Z_1, \dots, Z_n \overset{\text{iid}}{\sim} P_0$ (randomised trials, independently sampled survey responses, repeated coin tosses). It implies the LLN, the CLT and $\sqrt{n}$-convergence, and enables classical inference: confidence intervals, hypothesis tests, t-tests and regression inference.

4.2 Beyond i.i.d.

- **Dependence**: temperature evolves gradually; if today is hot, tomorrow probably is too. Independence would require $\mathrm{Cov}(Z_t, Z_{t+1}) = 0$. With positive autocorrelation the effective sample size is smaller than n: for a stationary AR(1) with correlation ρ, $\mathrm{Var}(\bar{Z}) \approx \frac{\sigma^2}{n}\frac{1+\rho}{1-\rho}$.
- **High dimension**: classically p fixed and $n \to \infty$; in modern data $p \gg n$. When $p > n$ the linear system $Y = X\beta$ is underdetermined and OLS is not unique, so additional structure is required, e.g. sparsity $\|\beta\|_0 = s \ll n$.
- **Curse of dimensionality**: the number of points needed to cover the space grows multiplicatively in the dimension; covering $[0,1]^p$ by cubes of side ϵ needs $N(\epsilon) \asymp (1/\epsilon)^p$ cubes. High dimension implies exponential complexity, and local methods (Ch. 2) need exponentially many observations.
- **Partial observation**: we do not always observe the full outcome. With survival time T and censoring time C we record $\tilde{T} = \min(T, C)$; the study may end before the event occurs.
- **Measurement error**: the observed predictor may differ from the true one, $\tilde{X} = X + \delta$, and the outcome contains noise, $Y = f(X) + \varepsilon$; observed relationships may be distorted (attenuation, Ch. 5).

Remark: understanding data means ...

knowing what is observed, what is assumed, what is unobserved, and the limits of generalisation.

Derivation: why the curse of dimensionality bites local methods

Let X be uniform on $[0,1]^p$. A cube neighbourhood that captures a fraction r of the data must have side length $e_p(r) = r^{1/p}$. For $r = 0.01$ and $p = 10$, $e_p = 0.01^{0.1} \approx 0.63$: a "local" neighbourhood covers 63% of the range of every coordinate, so it is no longer local. Equivalently, for n uniform points the median distance from the origin to the nearest point approaches $1/2$ as p grows.

5 Exploratory Data Analysis

5.1 Visualisation

EDA visualises the distribution of each variable, relationships between variables, structure (correlation, clusters), high-dimensional structure (projections), temporal and spatial patterns, anomalies and missing data, and model-based diagnostics.

5.2 The geometry of the data cloud

Gaussian distributions produce elliptical density contours determined by the covariance matrix. Introductory statistics studies the geometry of the cloud:

Aspect	Summary	Classical test
centre (location)	mean	z-test, t-test
dispersion (spread)	standard deviation	χ^2-test, F-test
orientation (direction)	correlation	t-test for correlation

Beyond centre and spread, higher moments describe the **shape**: skewness $\mathbb{E}[(X-\mu)^3]/\sigma^3$ (asymmetry) and kurtosis $\mathbb{E}[(X-\mu)^4]/\sigma^4$ (tail heaviness; 3 for the normal).

5.3 Other summaries

Quantiles (median, quartiles, percentiles), the five-number summary $(\min, Q_1, \text{median}, Q_3, \max)$, robust statistics (median, IQR $= Q_3 - Q_1$, MAD $= \text{median}\,|X_i - \text{median}\,X|$), and visual summaries (boxplots, density plots). Robust summaries have high breakdown points: one wild value can move the mean arbitrarily far but cannot move the median beyond the range of the remaining data.

Summary

- Statistical learning estimates $P(Y \mid X)$ (supervised) or the structure of $P(X)$ (unsupervised) from finite, noisy samples.
- Data are random evidence about a population; without assumptions the target is not identified.
- The model $Y = f(X) + \varepsilon$ with a chosen function class fixes the flexibility and the mathematical tools.
- Dependence, high dimension, censoring and measurement error change what can be learned; EDA examines centre, spread, orientation and shape before modelling.

Chapter 2. Supervised Learning and Model Assessment

Lecture 4 (regression function, parametric vs nonparametric, KNN, bias–variance, adjusted R^2, AIC, BIC)

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1 The Supervised Learning Setup

Example: advertising data

A marketing data set records product `Sales` and the advertising budgets `TV`, `Radio`, `Digital` in many markets. Goal: learn f with $\texttt{Sales} \approx f(\texttt{TV}, \texttt{Radio}, \texttt{Digital})$. Response $Y = \texttt{Sales}$; predictors $X = (X_1, X_2, X_3)$; model

$$Y = f(X) + \varepsilon,$$

with f the unknown regression function and ε noise or unobserved factors ($\mathbb{E}\varepsilon = 0$, independent of X). A second example: income of 30 individuals versus years of education, seniority and marital status — which variables truly explain income?

1.1 Why estimate f?

Prediction: use $\hat{f}$ to predict Y at a new $X = x$. **Inference**: understand how X influences Y — which predictors matter, and the direction and size of their effects (which media increase sales, which channel has the largest impact, what is the expected change in sales from an extra \$1,000 on TV?).

1.2 The regression function

At a given value, say $X = 4$, many Y values are possible. What is the best prediction?

Theorem: the conditional mean is the optimal predictor

$$f(x) = \arg\min_g \mathbb{E}[(Y - g(X))^2 \mid X = x] = \mathbb{E}(Y \mid X = x),$$

called the **regression function**. Moreover, for any g, $\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - f(X))^2] + \mathbb{E}[(f(X) - g(X))^2]$.

Proof. For a constant c, $\mathbb{E}[(Y - c)^2 \mid X = x] = \mathrm{Var}(Y \mid X = x) + (f(x) - c)^2$, minimised at $c = f(x)$. For the decomposition, write $Y - g = (Y - f) + (f - g)$; the cross term vanishes because $\mathbb{E}[(Y - f(X)) \mid X] = 0$. ■

The term $\mathrm{Var}(Y \mid X = x) = \mathrm{Var}(\varepsilon)$ is the **irreducible error**: even the true f cannot remove it. The term $\mathbb{E}(f - g)^2$ is the **reducible error** that learning tries to shrink.

2 Parametric Approach

Parametric methods impose structure on f:

1. specify a functional form, e.g. the linear model $f_L(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$;
2. estimate the parameters $(\beta_0, \ldots, \beta_p)$ from data (e.g. by least squares).

f is restricted to a low-dimensional parameter space: this gains interpretability and stability but risks **model misspecification**. Adding parameters increases flexibility: the linear model $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ versus the quadratic $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$; with two inputs (education, seniority) the linear model is a plane and a polynomial model a curved surface.

3 Nonparametric Approach

3.1 Local averaging

We rarely have many observations with exactly $X = 4$, so $\mathbb{E}(Y \mid X = x)$ cannot be computed directly. Instead we average nearby points:

$$\hat{f}(x) = \mathrm{Ave}(Y \mid X \in \mathcal{N}(x)),$$

with $\mathcal{N}(x)$ a neighbourhood of x.

Definition: K-nearest neighbours regression

$$\hat{f}(x) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(x)} y_i,$$

where $\mathcal{N}_K(x)$ is the set of the K training points closest to x.

Other nonparametric methods: kernel regression, smoothing splines (Ch. 7), Gaussian-process regression. With $K = 1$ the fit interpolates the data and is very rough; with $K = 9$ it is smoother.

Theorem: bias and variance of KNN at a point

Let $Y = f(X) + \varepsilon$ with $\mathrm{Var}\varepsilon = \sigma^2$ and condition on the inputs. With neighbours $x_{(1)}, \dots, x_{(K)}$ of x_0,

$$\mathbb{E}\big[(Y_0 - \hat{f}(x_0))^2\big] = \sigma^2 + \Big[f(x_0) - \frac{1}{K}\sum_{l=1}^{K} f(x_{(l)})\Big]^2 + \frac{\sigma^2}{K}.$$

Proof. $\hat{f}(x_0) = \frac{1}{K}\sum_l f(x_{(l)}) + \frac{1}{K}\sum_l \varepsilon_{(l)}$; its variance is σ^2/K and its mean is the average of f over the neighbours. Apply the bias–variance decomposition (§4.3). ■

Small K: low bias (neighbours are close), high variance σ^2/K. Large K: neighbours spread out, bias grows. K is the complexity parameter (*smaller* K = *more* flexible).

3.2 Pros and cons

Nonparametric methods make no explicit assumption about the form of f, giving flexible estimates that try to get close to the data without being too rough or wiggly. The price: they need a very large number of observations (and suffer from the curse of dimensionality, Ch. 1). An extremely flexible fit can reproduce the training data almost perfectly and still predict badly.

4 Assessing Model Accuracy

4.1 Complexity, underfitting, overfitting

Simple models impose strong structural assumptions (parametric, e.g. linear regression); nonparametric models (nearest neighbours, splines) are more flexible. **Underfitting**: the model is too simple, producing systematic errors. **Overfitting**: the model follows the training data too closely and fits noise rather than signal.

4.2 Training error and test error

For a model fitted on $\mathrm{Tr} = \{(x_i, y_i)\}_{i=1}^{n}$,

$$\mathrm{MSE}_{\mathrm{Tr}} = \mathrm{Ave}_{i \in \mathrm{Tr}} \big[y_i - \hat{f}(x_i)\big]^2, \qquad \mathrm{MSE}_{\mathrm{Te}} = \mathrm{Ave}_{i \in \mathrm{Te}} \big[y_i - \hat{f}(x_i)\big]^2,$$

with $\mathrm{Te} = \{(x_i, y_i)\}_{i=1}^{M}$ fresh observations. Training error is overly optimistic, especially for flexible models; what we care about is performance on new data. Three simulated cases illustrate it: (1) a moderately curved truth — training MSE falls monotonically with flexibility while test MSE is U-shaped; (2) a nearly linear truth — the linear model and a smooth fit do best; (3) a wiggly truth with low noise — very flexible fits do best.

4.3 Bias–variance trade-off

Theorem: decomposition of the expected test error

Let (x_0, y_0) be a new observation, $y_0 = f(x_0) + \varepsilon_0$, $f(x) = \mathbb{E}(Y \mid X = x)$, and let $\hat{f}$ be fitted on an independent training set. Then

$$\mathbb{E}\big[(y_0 - \hat{f}(x_0))^2\big] = \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\big[\mathrm{Bias}(\hat{f}(x_0))\big]^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\varepsilon)}_{\text{irreducible}} .$$

Proof. $y_0 - \hat{f} = \varepsilon_0 + (f - \mathbb{E}\hat{f}) + (\mathbb{E}\hat{f} - \hat{f})$; the three terms are uncorrelated (ε_0 is independent of the training data and has mean 0; the second term is constant; the third has mean 0). Square and take expectations. ■

As flexibility increases, bias decreases and variance increases. The challenge is to find a method for which *both* are low.

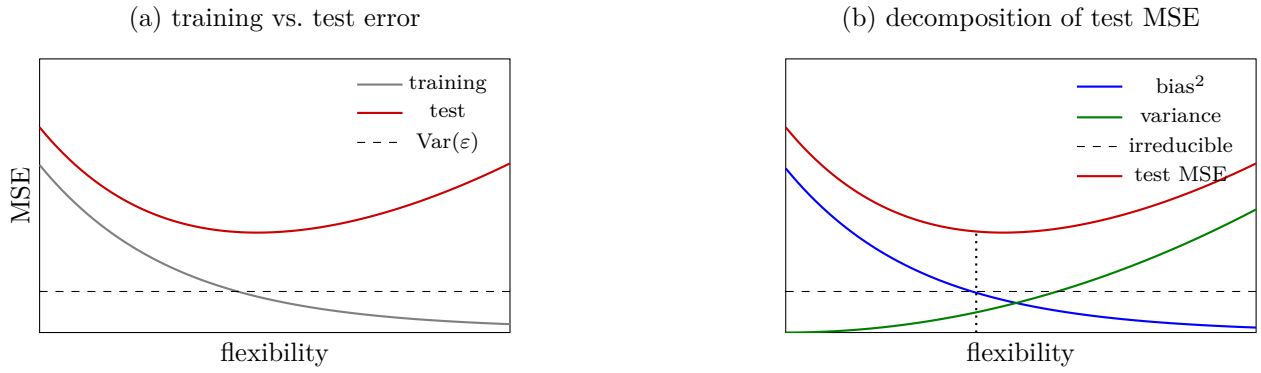


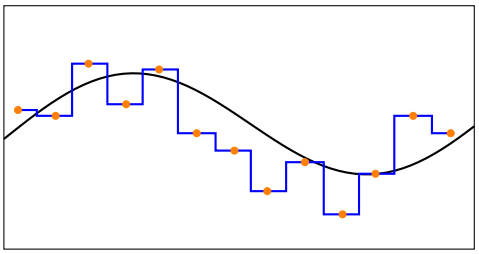
Figure 1: Schematic bias–variance trade-off. Training error always falls with flexibility; test error is U-shaped and never goes below the irreducible error. The dotted line marks the optimal flexibility.

5 Criteria Based on the Training Data

Besides test error, some criteria evaluate models on the training data by adding a complexity penalty:

$$\text{Score} = \text{goodness of fit} + \text{complexity penalty}, \qquad \mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

$K = 1$: low bias, high variance



$K = 9$: smoother, more bias

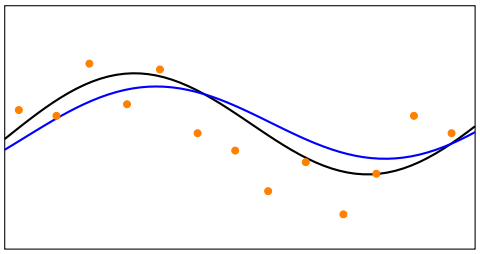


Figure 2: KNN regression in one dimension (schematic). Black: true f; blue: fit. $K = 1$ reproduces every training point (step function through the data); larger K averages more neighbours and flattens peaks and troughs.

5.1 Adjusted R^2

$R^2 = 1 - \text{RSS}/\text{TSS}$ always increases when a predictor is added (RSS cannot increase when the model is enlarged). The adjusted version penalises complexity:

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}.$$

Adding a predictor increases R^2_{adj} only if it lowers $\text{RSS}/(n-p-1)$, i.e. if it improves the fit sufficiently (equivalently, if its t-statistic exceeds 1 in absolute value).

5.2 AIC and BIC

For a Gaussian linear model with p predictors,

$$\text{AIC} = n\log(\text{RSS}/n) + 2(p+1), \qquad \text{BIC} = n\log(\text{RSS}/n) + \log(n)(p+1);$$

choose the model with the smallest value. In general, with k parameters, $\text{AIC} = -2\log L(\hat\theta) + 2k$ and $\text{BIC} = -2\log L(\hat\theta) + k\log n$. (For the Gaussian model $-2\log L(\hat\theta) = n\log(\text{RSS}/n) + \text{const.}$)

Derivation: information-theoretic interpretation of AIC

Measure model quality by the Kullback–Leibler divergence $\text{KL}(p_0 \| p_\theta) = \mathbb{E}_{p_0} \log \frac{p_0(X)}{p_\theta(X)}$; smaller is closer to the truth. Up to a constant this equals $-\mathbb{E}_{p_0} \log p_\theta(X)$, the expected log-loss on a *new* observation. The in-sample maximised log-likelihood $\log L(\hat\theta)$ is optimistic because $\hat\theta$ was fitted to the same data. A second-order Taylor expansion around the best parameter θ^* (correct model, regular case) shows that each of the two steps — fitting and evaluating on new data — contributes about $k/2$ in expectation, so

$$\mathbb{E}\big[-2\,\mathbb{E}_{\text{new}} \log p_{\hat\theta}(X^{\text{new}})\big] \approx \mathbb{E}\big[-2\log L(\hat\theta)\big] + 2k.$$

Hence $\text{AIC} = -2\log L(\hat\theta) + 2k$ estimates the expected out-of-sample deviance (Akaike).

Derivation: Bayesian interpretation of BIC

For candidate models $M_1, \dots, M_K$ with equal prior weights, Bayesian model selection picks the largest posterior $P(M_k \mid \text{data}) \propto \int L_k(\theta)\pi_k(\theta)d\theta$. A Laplace approximation of the integral around $\hat\theta_k$ gives

$$\log P(M_k \mid \text{data}) \approx \log L(\hat\theta_k) - \frac{k}{2}\log n + O(1)$$

(Schwarz, 1978). Multiplying by -2 gives BIC, so BIC approximates the Bayesian model evidence and the smallest BIC corresponds to the largest posterior probability.

	Penalty	Goal	Model size	Consistent?
AIC	$2k$	prediction	larger models	no
BIC	$k \log n$	true-model selection	smaller models	yes

AIC focuses on predictive performance; BIC prefers simpler models when n is large ($\log n > 2$ once $n \geq 8$). **Consistency**: if the true model is among the candidates, BIC selects it with probability $\to 1$, whereas AIC keeps a positive probability of overfitting. In practice: AIC (or CV) for prediction, BIC to identify the true structure.

Summary

- The regression function $\mathbb{E}(Y \mid X = x)$ is the optimal predictor under squared loss; $\mathrm{Var}(\varepsilon)$ is irreducible.
- Parametric methods assume a form for f (interpretable, may be misspecified); nonparametric methods such as KNN average locally (flexible, data-hungry).
- Training error is optimistic; test error is U-shaped in flexibility because of the bias–variance trade-off.
- Adjusted R^2, AIC and BIC penalise complexity; AIC targets prediction, BIC consistency.

Chapter 3. Resampling: Cross-Validation and the Bootstrap

Lecture 5 (validation set, LOOCV, K-fold CV, right and wrong CV, bootstrap SE and CI)

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1 What Are Resampling Methods?

Resampling methods repeatedly draw samples from a training set and refit the model of interest on each sample, to learn more about the fitted model:

- **model assessment**: estimate test error rates;
- **model selection**: choose the appropriate level of flexibility.

They are computationally expensive, but feasible with modern computers. Two methods: cross-validation and the bootstrap.

The best way to estimate prediction error is a large designated test set, which is often unavailable. AIC and BIC adjust the training error mathematically (Ch. 2). Here we instead hold out a subset of the training observations from fitting and evaluate on it.

2 Cross-Validation

2.1 Validation-set approach

Randomly split the samples into a training set and a validation set; fit on the training set, predict the validation set, and use the validation error (MSE for quantitative, misclassification rate for qualitative responses) as an estimate of the test error.

Limitations. (i) The estimate can be highly variable, depending on which observations fall in which set. (ii) Only part of the data is used for fitting, so the validation error tends to *overestimate* the test error of the model fitted on all data.

2.2 Leave-one-out cross-validation

For each model: split the n observations into a training set of size $n-1$ and a validation set of size 1; fit, compute $\mathrm{MSE}_i = (y_i - \hat{y}_{(i)})^2$; repeat n times and average,

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i.$$

LOOCV has less bias than the validation-set approach (each fit uses almost all data) and no randomness in the split, but it requires n fits.

Theorem: LOOCV shortcut for least squares and linear smoothers

If $\hat{y} = Hy$ for a fixed matrix H with the leave-one-out property (true for least squares, ridge, smoothing splines), then

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2,$$

so LOOCV costs a single fit.

Proof for least squares. By the Sherman–Morrison formula, removing row i changes the estimate by $\hat{\beta} - \hat{\beta}_{(i)} = (X^\top X)^{-1} x_i e_i/(1-h_{ii})$ with $e_i = y_i - \hat{y}_i$. Hence $y_i - x_i^\top \hat{\beta}_{(i)} = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$. ■

Points with high leverage h_{ii} have their residuals inflated, as they should be.

2.3 K-fold cross-validation

Randomly divide the data into K (roughly) equal parts $C_1, \ldots, C_K$ with n_k observations in part k ($n_k = n/K$ if K divides n). Leave out part k, fit on the remaining $K-1$ parts, and predict part k; do this for every k and combine:

$$\mathrm{CV}_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k, \qquad \mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in C_k} (y_i - \hat{y}_i)^2,$$

where $\hat{y}_i$ is fitted from the data with part k removed. $K = n$ gives LOOCV. The estimate is used both to select the best model and to give an idea of the test error of the chosen model.

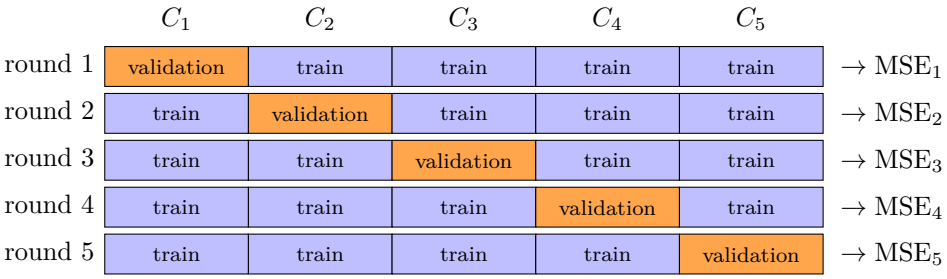


Figure 1: 5-fold cross-validation. Each part is held out once; the CV estimate averages $\mathrm{MSE}_1, \ldots, \mathrm{MSE}_5$. LOOCV is the extreme case with n parts of size one.

2.4 K-fold CV versus LOOCV

Typical choices are $K = 5$ or 10. The obvious advantage of K-fold CV is computational. Less obvious: LOOCV "does not shake up the data enough" — the n fitted models are trained on nearly identical data, so their errors are *highly correlated* and their average can have high variance. K-fold CV has somewhat more bias (training sets of size $(K-1)n/K$) but lower variance: a bias–variance trade-off for the error estimate itself. On the Auto data (mpg versus polynomials in horsepower), the LOOCV curve and nine repetitions of 10-fold CV with different random splits give very similar curves.

2.5 Cross-validation: right and wrong

Example: selecting predictors before CV

With 5000 predictors and 200 samples: (1) select the 100 predictors most correlated with the response; (2) fit a model using only these 100. Can we apply CV only to step 2? **No.** Step 1 has already used the responses of the entire data set, so the validation folds are not independent of the selection. Even if the predictors are pure noise, the selected 100 look predictive on all folds and the CV error is optimistically biased (in the classic simulation it is near 0 although the true error is 50%). The right way applies CV to *steps 1 and 2 together*: in each fold, select the predictors using only the training part.

Feature selection is a form of training and must be included in the validation process.

2.6 What does CV estimate?

For a model $\hat{f}$ trained on data, the prediction risk is $R(\hat{f}) = \mathbb{E}[(Y - \hat{f}(X))^2]$; new data are unavailable, so CV approximates the expected prediction error by repeatedly splitting into training and validation parts. **CV estimates the prediction risk.**

Theory and limitation. Stone (1977) showed that LOOCV is asymptotically equivalent to AIC; both aim to minimise prediction risk, which justifies CV for prediction. But CV does not necessarily identify the true model: Shao (1997) showed that LOOCV may prefer overly complex models (it is not consistent for model selection, like AIC). Takeaway: prediction → CV, AIC; true-model identification → BIC.

3 The Bootstrap

The bootstrap is a flexible and powerful tool to quantify the uncertainty of an estimator or a learning method, e.g. the standard error of a coefficient or a confidence interval.

3.1 A motivating example

We invest a fraction α of our money in an asset with return X and $1-\alpha$ in an asset with return Y, and want to minimise the risk $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

Theorem: minimum-variance allocation

With $\sigma_X^2 = \mathrm{Var}X$, $\sigma_Y^2 = \mathrm{Var}Y$, $\sigma_{XY} = \mathrm{Cov}(X,Y)$,

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof. $V(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$ is a convex quadratic; $V'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0$ gives $\alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}$. ■

The variances are unknown, so we plug in estimates:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}.$$

How accurate is $\hat{\alpha}$? In a simulation (true $\alpha = 0.6$), four data sets of 100 pairs gave $\hat{\alpha} = 0.576, 0.532, 0.657, 0.651$. Repeating 1,000 times gave $\bar{\alpha} = \frac{1}{1000}\sum_r \hat{\alpha}_r = 0.5996$ and standard deviation $\sqrt{\frac{1}{999}\sum_r(\hat{\alpha}_r - \bar{\alpha})^2} = 0.083$, so $\mathrm{SE}(\hat{\alpha}) \approx 0.083$.

3.2 Back to the real world

For real data we cannot draw new samples from the population. The bootstrap uses the computer to *mimic* this: instead of independent data sets from the population, it repeatedly samples observations **from the original data set with replacement**, each bootstrap data set having the same size n. Some observations appear several times, others not at all. In the portfolio example, histograms of 1,000 estimates from true resampling and from 1,000 bootstrap samples of one data set look very similar.

Algorithm: bootstrap standard error

For $b = 1, \ldots, B$ (say $B = 100$ or 1000): draw Z^{*b} by sampling n observations with replacement, and compute $\hat{\alpha}^{*b}$. Then

$$\mathrm{SE}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}\right)^2}$$

estimates the standard error of $\hat{\alpha}$ computed from the original data.

Why it works. The bootstrap replaces the unknown P_0 by the empirical distribution $\hat{P}_n$ (mass $1/n$ at each observation) — the *plug-in principle*. Sampling from $\hat{P}_n$ mimics sampling from P_0, and for smooth statistics the bootstrap distribution of $\hat{\alpha}^* - \hat{\alpha}$ approximates that of $\hat{\alpha} - \alpha$.

3.3 Bootstrap in general

More complex data need adapted schemes. For time series we cannot resample observations independently; the **block bootstrap** resamples blocks of consecutive observations with replacement and pastes them together, preserving short-range dependence.

3.4 Confidence intervals

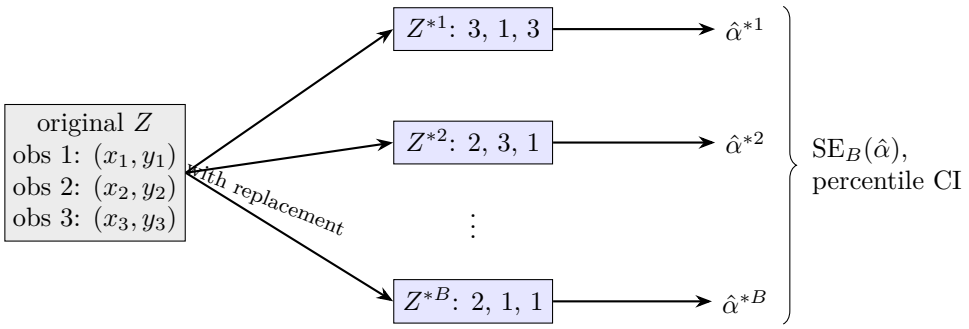


Figure 2: The bootstrap with $n = 3$. Each bootstrap set has n observations drawn with replacement (some repeated, some missing); the spread of the B estimates approximates the sampling distribution of $\hat{\alpha}$.

Besides standard errors, the bootstrap gives approximate confidence intervals. From the histogram of 1,000 bootstrap $\hat{\alpha}^*$, the 5% and 95% quantiles are $(0.43, 0.72)$: an approximate 90% **bootstrap percentile interval** for α. Alternatives: the normal interval $\hat{\alpha} \pm z_{1-\alpha/2}\mathrm{SE}_B$, and the basic interval $(2\hat{\alpha} - q_{0.95},\ 2\hat{\alpha} - q_{0.05})$.

3.5 The wrong way: bootstrap prediction error

One might use each bootstrap sample as a training set and the original sample as the validation set. In CV, validation folds are disjoint from training folds; but each bootstrap sample overlaps heavily with the original data: an observation appears in a given bootstrap sample with probability $1 - (1 - 1/n)^n \approx 1 - e^{-1} \approx 0.632$. This seriously *underestimates* the true prediction error. (Using only out-of-bag observations for each bootstrap fit fixes this, leading to the ".632" estimators and to OOB error in bagging, Ch. 11.)

Summary

- Resampling reuses the observed data for two goals: estimating prediction error and quantifying the uncertainty of an estimator.
- The validation-set estimate is variable and biased upward; LOOCV is nearly unbiased but its fold errors are highly correlated; $K = 5$ or 10 balances both.
- Every data-dependent step (including variable selection) must be inside the CV loop.
- CV is asymptotically like AIC (prediction), not like BIC (identification).
- The bootstrap resamples with replacement to estimate standard errors and confidence intervals; it is not a substitute for CV in estimating prediction error.

- Prediction → cross-validation; uncertainty → bootstrap.

Chapter 4. Linear Regression

Lecture 6-1 (simple and multiple regression, inference, F-test, qualitative predictors, Gauss–Markov, interactions, polynomials)

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1 Simple Linear Regression

Linear regression assumes that the dependence of Y on $X_1, \ldots, X_p$ is linear. True regression functions are never exactly linear, yet the model is extremely useful both conceptually and practically. On the advertising data we may ask: is there a relationship between budget and sales? How strong is it? Which media contribute?

1.1 Model and least squares

$$Y = \beta_0 + \beta_1 X + \varepsilon, \qquad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

β_0 (intercept) and β_1 (slope) are the coefficients. With residuals $e_i = y_i - \hat{y}_i$, the residual sum of squares is $\mathrm{RSS} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$.

Theorem: least-squares estimates

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Proof. $\partial \mathrm{RSS}/\partial \beta_0 = -2\sum(y_i - \beta_0 - \beta_1 x_i) = 0$ gives $\beta_0 = \bar{y} - \beta_1 \bar{x}$. Substituting, $\partial \mathrm{RSS}/\partial \beta_1 = -2\sum(x_i - \bar{x})\{(y_i - \bar{y}) - \beta_1(x_i - \bar{x})\} = 0$ (using $\sum(x_i - \bar{x}) = 0$) gives $\hat{\beta}_1$. RSS is a convex quadratic, so this is the minimum. ■

On the advertising data, $\widehat{\texttt{sales}} = 7.03 + 0.0475 \times \texttt{TV}$; the line captures the essence of the relationship although it overestimates the trend at the left of the plot.

1.2 Accuracy of the coefficient estimates

With a true line $f(X) = 2 + 3X$, least-squares lines computed from different random samples scatter around it. The standard error measures this variation under repeated sampling:

Theorem: standard errors

If the errors are uncorrelated with common variance $\sigma^2 = \mathrm{Var}(\varepsilon)$ and the x_i are fixed,

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \qquad \mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \Big[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\Big].$$

Proof. $\hat{\beta}_1 = \sum_i w_i y_i$ with $w_i = (x_i - \bar{x})/S_{xx}$, $S_{xx} = \sum(x_i - \bar{x})^2$; so $\mathrm{Var}\hat{\beta}_1 = \sigma^2 \sum w_i^2 = \sigma^2/S_{xx}$. Next $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\mathrm{Cov}(\bar{y}, \hat{\beta}_1) = \frac{\sigma^2}{n}\sum w_i = 0$, so $\mathrm{Var}\hat{\beta}_0 = \sigma^2/n + \bar{x}^2\sigma^2/S_{xx}$. ■

σ is estimated by the **residual standard error** $\mathrm{RSE} = \sqrt{\mathrm{RSS}/(n-2)}$. A spread-out design (large S_{xx}) gives a more precise slope.

1.3 Confidence intervals and tests

A $100(1-\alpha)\%$ confidence interval is a range that, under repeated sampling, contains the true parameter with probability $1 - \alpha$:

$$\hat{\beta}_1 \pm t_{1-\alpha/2,\, n-2} \cdot \mathrm{SE}(\hat{\beta}_1) \quad (\approx \hat{\beta}_1 \pm 2\,\mathrm{SE}(\hat{\beta}_1) \text{ for } 95\%).$$

For the advertising data the 95% interval for β_1 is $[0.042, 0.053]$.

Testing H_0: no relationship between X and Y ($\beta_1 = 0$, so $Y = \beta_0 + \varepsilon$) against H_A: $\beta_1 \neq 0$ uses

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)} \sim t_{n-2} \text{ under } H_0 \text{ (Gaussian errors)},$$

and the p-value is the probability of observing $|t|$ or larger.

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

1.4 Overall accuracy: RSE and R^2

$\mathrm{RSE} = \sqrt{\frac{1}{n-2}\sum(y_i - \hat{y}_i)^2}$ estimates the standard deviation of ε and measures lack of fit (in the units of Y).

$$R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad \mathrm{TSS} = \sum(y_i - \bar{y})^2,$$

the fraction of variance explained, between 0 and 1.

Theorem: $R^2 = r^2$ in simple regression

With $r = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}}$, simple linear regression has $R^2 = r^2$.

Proof. $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, and the decomposition $\mathrm{TSS} = \mathrm{RSS} + \sum(\hat{y}_i - \bar{y})^2$ holds (normal equations). So $R^2 = \hat{\beta}_1^2 S_{xx}/S_{yy} = S_{xy}^2/(S_{xx}S_{yy}) = r^2$. ■

For Sales on TV: RSE = 3.26, $R^2 = 0.612$, $F = 312.1$. Actual sales deviate from the true line by about 3,260 units on average; about two thirds of the variability in sales is explained by TV. What counts as a good R^2 depends on the application.

2 Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon, \qquad \texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{radio} + \beta_3 \texttt{newspaper} + \varepsilon.$$

β_j is the average effect on Y of a one-unit increase in X_j, *holding all other predictors fixed.*

2.1 Interpreting coefficients

Correlations among predictors cause problems: interpretations become hazardous (when X_j changes, everything else changes) and the variances of all coefficients tend to increase, sometimes dramatically. The ideal is uncorrelated predictors (a balanced design), where each coefficient can be estimated and tested separately. Claims of causality should be avoided for observational data.

Example: the woes of interpreting coefficients (Mosteller and Tukey, 1977)

Y = total money in your pocket, X_1 = total number of coins, X_2 = number of pennies, nickels and dimes. Alone, X_2 is positively associated with Y. But given X_1, more small coins means fewer quarters, so the coefficient of X_2 becomes negative.

2.2 Estimation

The $\hat{\beta}_j$ minimise RSS $= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip})^2$; in matrix form $\hat{\beta} = (X^\top X)^{-1} X^\top y$ (normal equations $X^\top (y - X\hat{\beta}) = 0$), and $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$. With $p = 2$ the fit is the plane minimising squared vertical distances.

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

For given TV and newspaper budgets, an extra \$1,000 on radio is associated with about 189 additional units sold. The newspaper coefficient is near zero although newspaper alone is significant in a simple regression: newspaper is correlated with radio and acts as a surrogate. Simple and multiple regression coefficients can differ greatly.

3 Four Important Questions

3.1 Is at least one predictor useful?

Test $H_0 : \beta_1 = \cdots = \beta_p = 0$ against H_A: at least one $\beta_j \neq 0$, with

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)} \;\sim\; F_{p,\,n-p-1} \text{ under } H_0.$$

Derivation: why F is the right statistic

Under the linear model, $\mathbb{E}[\mathrm{RSS}/(n-p-1)] = \sigma^2$, and under H_0 also $\mathbb{E}[(\mathrm{TSS}-\mathrm{RSS})/p] = \sigma^2$; if H_0 is false the numerator is larger. With Gaussian errors, $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p-1}$ and $(\mathrm{TSS} - \mathrm{RSS})/\sigma^2 \sim \chi^2_p$ independently under H_0 (Cochran), giving the F distribution. Individual t-tests are not enough: with $p = 100$ and H_0 true, about 5 coefficients will have $p < 0.05$ by chance, while the F-test keeps its level.

Advertising data: RSE $= 1.69$, $R^2 = 0.897$, $F = 570$.

3.2 Deciding on the important variables

Best subsets: fit all subsets and choose by a criterion that balances training error with model size; infeasible for large p (2^p models; over a billion for $p = 40$). Automated searches: **forward selection** starts with no predictors and adds, one at a time, the variable giving the greatest improvement; **backward elimination** starts with the full model and removes the least useful predictor one at a time. Each searches only $1 + p(p+1)/2$ models (details in Ch. 6).

3.3 How well does the model fit?

RSE $= \sqrt{\mathrm{RSS}/(n-p-1)}$, $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$, and visualisation of residuals. Note R^2 never decreases when predictors are added, while RSE can increase if the decrease of RSS is small relative to the lost degree of freedom.

3.4 How accurate is the prediction?

Three sources of uncertainty: (i) **reducible error** — $\hat{Y} = \hat{\beta}_0 + \cdots + \hat{\beta}_p X_p$ only estimates $f(X) = \beta_0 + \cdots + \beta_p X_p$ (a confidence interval quantifies it); (ii) **model bias** — the linear model is an approximation; (iii) **irreducible error** ε — how much will Y vary from $\hat{Y}$ (a prediction interval, always wider, includes it). For a new x_0: $\mathrm{Var}(x_0^\top \hat{\beta}) = \sigma^2 x_0^\top (X^\top X)^{-1} x_0$ and $\mathrm{Var}(Y_0 - x_0^\top \hat{\beta}) = \sigma^2\{1 + x_0^\top (X^\top X)^{-1} x_0\}$.

4 Qualitative Predictors

Some predictors take a discrete set of values (categorical predictors, factors), e.g. student status or marital status in the Credit data.

4.1 Two levels

For the difference in credit-card balance between house owners and non-owners, define $x_i = 1$ if person i owns a house and 0 otherwise:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{owner,} \\ \beta_0 + \varepsilon_i & \text{non-owner.} \end{cases}$$

Estimates: intercept 509.80 (SE 33.13), own[Yes] 19.73 (SE 46.05, $p = 0.669$). Average debt is \$509.80 for non-owners and \$529.53 for owners; the difference is not significant.

4.2 More than two levels

For region (East, South, West) create two dummies $x_{i1} = \mathbb{I}(\text{South})$, $x_{i2} = \mathbb{I}(\text{West})$:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{South,} \\ \beta_0 + \beta_2 + \varepsilon_i & \text{West,} \\ \beta_0 + \varepsilon_i & \text{East (baseline).} \end{cases}$$

There is always one fewer dummy than levels (with an intercept, K dummies would be collinear with the intercept column). Estimates: baseline East \$531.00; South -18.69 ($p = 0.774$); West -12.50 ($p = 0.826$): no evidence of a regional difference. The coefficients depend on the baseline, but the fitted values and the overall F-test do not.

5 The Gauss–Markov Theorem

Theorem: Gauss–Markov

Let $y = X\beta + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}\varepsilon = \sigma^2 I$, and X of full column rank. For any fixed c, among all linear unbiased estimators of $\theta = c^\top \beta$, the estimator $\hat{\theta} = c^\top \hat{\beta}$, $\hat{\beta} = (X^\top X)^{-1} X^\top y$, has the minimum variance, and it is unique. Least squares is the **BLUE**.

Proof. *Step 1 (unbiasedness constraint).* A linear estimator is $\tilde{\theta} = a^\top y$. Since $\mathbb{E}(a^\top y) = a^\top X\beta$, unbiasedness for all β requires $a^\top X\beta = c^\top \beta$ for all β, i.e. $X^\top a = c$. *Step 2 (variance comparison).* Least squares corresponds to $a_0 = X(X^\top X)^{-1} c$, which satisfies $X^\top a_0 = c$. Any other unbiased a can be written $a = a_0 + d$ with $X^\top d = 0$. Then $a_0^\top d = c^\top (X^\top X)^{-1} X^\top d = 0$, so

$$\mathrm{Var}(\tilde{\theta}) = \sigma^2 a^\top a = \sigma^2 (a_0^\top a_0 + d^\top d) \;\geq\; \sigma^2 a_0^\top a_0 = \mathrm{Var}(\hat{\theta}),$$

with equality iff $d = 0$. ■

Gauss–Markov needs neither normality nor independence (only uncorrelated, homoscedastic errors), but it restricts to *linear unbiased* estimators: biased estimators such as ridge (Ch. 6) can have smaller mean squared error.

6 Extensions of the Linear Model

The standard model assumes the effects are **additive** and **linear**.

6.1 Interactions

The additive model says that the effect of one more unit of TV is always β_1, regardless of radio. If radio spending makes TV advertising more effective (a *synergy* effect in marketing, an *interaction* in statistics), splitting a \$100,000 budget may beat spending it all on one medium. On the advertising data, residuals show that sales are overestimated when either budget is low and underestimated when it is split.

$$\texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{radio} + \beta_3 (\texttt{radio} \times \texttt{TV}) + \varepsilon.$$

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

R^2 rises from 89.7% to 96.8%: $(96.8-89.7)/(100-89.7) = 69\%$ of the variability left by the additive model is explained by the interaction. An extra \$1,000 of TV is associated with $(\hat\beta_1 + \hat\beta_3\,\texttt{radio}) \times 1000 = 19 + 1.1\,\texttt{radio}$ units; an extra \$1,000 of radio with $29 + 1.1\,\texttt{TV}$ units.

Hierarchy principle. If an interaction is included, include the main effects too, even if their p-values are not significant: interactions are hard to interpret without main effects, and omitting them changes the meaning of the interaction (and makes the model depend on the origin of the predictors).

6.2 Qualitative–quantitative interactions

For balance on income and student status,

$$\texttt{balance}_i \approx \beta_0 + \beta_1 \texttt{income}_i + \begin{cases} \beta_2 + \beta_3 \texttt{income}_i & \text{student} \\ 0 & \text{non-student} \end{cases}$$

$$= \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3)\texttt{income}_i & \text{student,} \\ \beta_0 + \beta_1 \texttt{income}_i & \text{non-student :} \end{cases}$$

different intercepts *and* slopes for the two groups.

6.3 Nonlinear effects: polynomial regression

On the Auto data, mpg versus horsepower is curved, and $\texttt{mpg} = \beta_0 + \beta_1 \texttt{hp} + \beta_2 \texttt{hp}^2 + \varepsilon$ fits better: estimates 56.9001, -0.4662, 0.0012, all with $p < 0.0001$. It is still a *linear* model (linear in the coefficients), fitted by least squares with $\texttt{hp}^2$ as an extra predictor. Residuals versus fitted values show a strong U-shape for the linear fit (nonlinearity) and little pattern for the quadratic fit.

Summary

- OLS minimises RSS; standard errors, t-tests and confidence intervals quantify the uncertainty of coefficients; RSE and R^2 measure fit.
- The F-test checks whether any predictor is useful; subset searches decide which.
- Dummy variables encode qualitative predictors relative to a baseline.
- Gauss–Markov: OLS is the best linear unbiased estimator.
- Interactions (with the hierarchy principle) and polynomial terms relax additivity and linearity while staying within least squares.

Chapter 5. Potential Problems in Regression

Lecture 6-2 (assumptions, non-Gaussianity, outliers and robust regression, breakdown point, measurement error)

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1 Assumptions and Common Issues

The linear model relies on: linearity; normally distributed errors; independent errors; constant variance; no outliers or high-leverage points; small collinearity.

Outliers are observations with unusually large residuals; **high-leverage points** have unusual predictor values; **correlated errors** violate independence.

Kind of problem	Examples
data problems	outliers, measurement error, missing data
model problems	omitted-variable bias, misspecification, multicollinearity
error problems	non-normality, heteroscedasticity, autocorrelation

Remark: diagnostics in brief

Nonlinearity: residuals versus fitted values show a pattern (fix: transformations, polynomials, splines). **Heteroscedasticity**: a funnel in the residual plot (fix: transform Y, e.g. $\log Y$ or $\sqrt{Y}$, or weighted least squares). **Correlated errors**: tracking in residuals over time; standard errors become too small. **Leverage**: $h_{ii} = x_i^\top (X^\top X)^{-1} x_i$, average $(p+1)/n$; flag $h_{ii} \gg (p+1)/n$. **Collinearity**: variance inflation factor $\mathrm{VIF}(\hat{\beta}_j) = 1/(1 - R^2_{X_j | X_{-j}})$; values above 5–10 are problematic.

2 Non-Gaussianity: Normality Is Not an Issue

Theorem: exact distribution under Gaussian errors

If $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I)$ and X of full column rank, then

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^\top X)^{-1}), \qquad \hat{\beta}_j \sim N(\beta_j, \sigma^2 v_j),$$

where v_j is the jth diagonal element of $(X^\top X)^{-1}$. Replacing σ by $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-p)}$ (p columns of X including the intercept),

$$t_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{v_j}} \sim t_{n-p},$$

which gives tests and confidence intervals.

Proof. $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$ is a linear map of a Gaussian vector, with mean β and covariance $\sigma^2 (X^\top X)^{-1}$. The residual vector $(I - H)\varepsilon$ is independent of $\hat{\beta}$ (uncorrelated jointly Gaussian), and $\mathrm{RSS}/\sigma^2 = \varepsilon^\top (I-H)\varepsilon/\sigma^2 \sim \chi^2_{n-p}$ since $I - H$ is a projection of rank $n - p$. The ratio of a standard normal and an independent $\sqrt{\chi^2_{n-p}/(n-p)}$ is t_{n-p}. ∎

2.1 Large-sample inference

If the errors are not normal, $\hat{\beta}$ is not exactly normal, but by the CLT

$$\hat{\beta} \;\dot{\sim}\; N(\beta, \sigma^2 (X^\top X)^{-1}) \quad \text{for large } n,$$

(provided errors are i.i.d. with finite variance and the design is not dominated by a few points, i.e. $\max_i h_{ii} \to 0$). Hence t-tests and confidence intervals remain approximately valid. Alternatively use

bootstrap inference: resample the data with replacement, recompute $\hat{\beta}$ for each bootstrap sample, and use the empirical distribution of $\hat{\beta}^*$ for standard errors and intervals (Ch. 3). Resampling pairs (x_i, y_i) is also valid under heteroscedasticity.

3 Outliers and Robust Regression

3.1 Why OLS is sensitive

OLS minimises $\sum_{i=1}^{n}(y_i - x_i^\top \beta)^2$. Squaring residuals amplifies large errors, so a single extreme observation can strongly influence the fitted model. This motivates robust regression.

3.2 Alternative objectives

- **Least absolute deviation (LAD)**: $\hat{\beta} = \arg\min_\beta \sum_i |y_i - x_i^\top \beta|$; the conditional median (Ch. 2 analogue: absolute loss is minimised by the median).
- **Huber regression**: $\hat{\beta} = \arg\min_\beta \sum_i \rho_c(r_i)$, $r_i = y_i - x_i^\top \beta$,

$$\rho_c(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le c, \\ c|r| - \frac{1}{2}c^2, & |r| > c. \end{cases}$$

- **Least trimmed squares (LTS)**: $\hat{\beta} = \arg\min_\beta \sum_{i=1}^{h} r^2_{(i)}$, using only the h smallest squared residuals ($r^2_{(1)} \le r^2_{(2)} \le \cdots$).

Derivation: Huber as iteratively reweighted least squares

ρ_c is convex with derivative $\psi_c(r) = r$ for $|r| \le c$ and $c\,\mathrm{sign}(r)$ otherwise. The estimating equation $\sum_i \psi_c(r_i)x_i = 0$ equals the weighted normal equation $\sum_i w_i r_i x_i = 0$ with $w_i = \min(1, c/|r_i|)$, so one iterates $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$. Large residuals get weights $c/|r_i| < 1$: their influence is bounded by c, whereas under OLS it is unbounded. Usually $c = 1.345\hat{s}$ for a robust scale $\hat{s}$ (95% efficiency under normal errors).

3.3 Breakdown point

Definition: breakdown point

The (finite-sample) breakdown point of an estimator is the smallest fraction of contaminated observations that can make the estimate arbitrarily large (break it down).

OLS	LAD	Huber M-estimator	least median of squares	least trimmed squares
0	0	0	0.5	≈ 0.5

OLS can be destroyed by a single outlier (breakdown $1/n \to 0$): moving one y_i to infinity moves $\hat{\beta}$ to infinity. LAD and Huber resist outliers in y, but a single bad *leverage point* (outlying x_i) can still break them, hence breakdown 0. High-breakdown estimators (LMS: minimise the median squared residual; LTS with $h \approx n/2$) tolerate nearly 50% contamination — more than half bad points can never be distinguished from good ones, so 0.5 is the maximum.

4 Measurement Error

Suppose the true model is $Y = \beta_0 + \beta_1 X + \varepsilon$, but the predictor is measured with error:

$$\tilde{X} = X + u, \qquad u \text{ independent of } (X, \varepsilon),\ \mathbb{E}u = 0.$$

Regressing Y on the observed $\tilde{X}$ gives a biased estimator.

Theorem: attenuation bias

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \frac{\mathrm{Var}(X)}{\mathrm{Var}(X) + \mathrm{Var}(u)} = \lambda \beta_1, \qquad 0 < \lambda \leq 1.$$

Proof. The OLS slope converges to $\mathrm{Cov}(\tilde{X}, Y)/\mathrm{Var}(\tilde{X})$. Here $\mathrm{Cov}(X + u, \beta_0 + \beta_1 X + \varepsilon) = \beta_1 \mathrm{Var} X$ and $\mathrm{Var}(\tilde{X}) = \mathrm{Var} X + \mathrm{Var} u$. ■

The slope is shrunk toward zero (**attenuation bias**); λ is the reliability ratio. Measurement error in Y only, in contrast, inflates the error variance but does not bias the slope. In multiple regression, error in one predictor also biases the coefficients of correlated predictors.

4.1 Remedies

Method	Idea
instrumental variables	use a variable Z correlated with X but independent of the measurement error: $\hat{\beta}_1^{\mathrm{IV}} = \widehat{\mathrm{Cov}}(Z, Y)/\widehat{\mathrm{Cov}}(Z, \tilde{X})$
repeated measurements	average m noisy measurements, reducing $\mathrm{Var} u$ to $\mathrm{Var} u/m$; replicates also estimate $\mathrm{Var} u$ so that $\hat{\beta}_1/\hat{\lambda}$ corrects the bias
errors-in-variables models	model the measurement process explicitly (e.g. Deming regression)
SIMEX	add extra artificial noise of increasing size, observe how the estimate changes, and extrapolate back to zero measurement error
validation data	use a subset where the true X is observed to calibrate $\tilde{X}$ (regression calibration)

Summary

- Regression problems arise from the data (outliers, measurement error, missing data), the model (omitted variables, misspecification, collinearity) and the error structure (heteroscedasticity, autocorrelation).
- Normality is often not critical: the CLT gives large-sample inference and the bootstrap is an alternative.
- Outliers motivate robust regression (LAD, Huber, LTS); the breakdown point measures robustness, and only high-breakdown methods resist leverage points.
- Error in a predictor causes attenuation bias toward zero; IV, replicates, SIMEX or validation data correct it.

Chapter 6. Explainable Models: Subset Selection, Shrinkage and Dimension Reduction

Lecture 6-3 (best subset, stepwise, C_p/AIC/BIC/CV, ridge, lasso, high-dimensional theory, PCR, PLS)

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1 Three Classes of Methods

- **Subset selection**: identify a subset of the p predictors related to the response and fit least squares on it.
- **Shrinkage (regularisation)**: fit all p predictors but shrink the coefficients toward zero relative to least squares; this reduces variance and can also select variables.
- **Dimension reduction**: project the p predictors onto an M-dimensional subspace ($M < p$) and regress on the M projections by least squares.

2 Subset Selection

2.1 Best subset selection

Algorithm: best subset selection

1. Let $\mathcal{M}_0$ be the null model (no predictors; predicts the sample mean).
2. For $k = 1, \ldots, p$: fit all $\binom{p}{k}$ models with exactly k predictors and call the best (smallest RSS, or largest R^2) $\mathcal{M}_k$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using a chosen score (CV error, C_p, AIC, BIC, adjusted R^2).

Within step 2, models of the same size can be compared by RSS; across sizes (step 3), RSS always favours the largest model, so a score that estimates test error is needed. Best subset examines 2^p models: accurate but computationally infeasible for large p, and statistically prone to overfitting (the huge search space lets a model fit noise).

2.2 Stepwise selection

Algorithm: forward stepwise selection

1. Let $\mathcal{M}_0$ be the null model.
2. For $k = 0, \ldots, p-1$: consider the $p-k$ models that add one predictor to $\mathcal{M}_k$; the best of them is $\mathcal{M}_{k+1}$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ by CV error, C_p, AIC, BIC or adjusted R^2.

Algorithm: backward elimination

1. Let $\mathcal{M}_p$ be the full model.
2. For $k = p, p-1, \ldots, 1$: consider the k models that drop one predictor from $\mathcal{M}_k$; the best is $\mathcal{M}_{k-1}$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$.

Both search $1 + \sum_{k=0}^{p-1}(p-k) = 1 + p(p+1)/2$ models (e.g. 211 instead of $2^{20} \approx 10^6$ for $p = 20$). Neither is guaranteed to find the best subset: the best two-variable model need not contain the

best one-variable predictor. Backward elimination needs $n > p$ so the full model can be fitted ($\hat{\beta} = (X^\top X)^{-1} X^\top Y$ requires $X^\top X$ invertible); forward selection works even with $n < p$ (up to $\mathcal{M}_{n-1}$) and is often the only viable subset method when p is very large.

2.3 Choosing the optimal model

Each procedure returns a sequence $\mathcal{M}_0, \dots, \mathcal{M}_p$; we choose the size $\hat{k}$. Indirect estimates of test error adjust the training error:

$$C_p = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2), \qquad \text{AIC, BIC (Ch. 2)}, \qquad R^2_{\text{adj}},$$

with d the number of parameters and $\hat{\sigma}^2$ estimated from the full model.

Theorem: C_p is unbiased for the in-sample prediction error

For a least-squares fit $\hat{y} = Hy$ with $\mathrm{tr}H = d$, $y = \mu + \varepsilon$, $\mathrm{Var}\varepsilon = \sigma^2 I$, and new responses y^{new} at the same inputs,

$$\mathbb{E}\Big[\frac{1}{n}\|y^{\text{new}} - \hat{y}\|^2\Big] = \mathbb{E}\Big[\frac{1}{n}\|y - \hat{y}\|^2\Big] + \frac{2d\sigma^2}{n}.$$

Proof. Expanding both sides, the difference is $\frac{2}{n}\sum_i \mathrm{Cov}(y_i, \hat{y}_i) = \frac{2}{n}\mathrm{trCov}(y, Hy) = \frac{2}{n}\sigma^2 \mathrm{tr}H = \frac{2d\sigma^2}{n}$. ■

The direct alternative is the validation-set or CV error for each $\mathcal{M}_k$: it estimates test error directly, makes fewer assumptions about the true model, and does not need $\hat{\sigma}^2$. On the Credit data, $\sqrt{\text{BIC}}$, validation error and CV error all flatten after about four to six predictors. A **one-standard-error rule** picks the smallest model whose estimated test error is within one SE of the minimum.

3 Shrinkage Methods

Instead of least squares on a subset, fit all p predictors with a technique that constrains or regularises the coefficients, shrinking them toward zero. Shrinking can significantly reduce their variance.

3.1 Ridge regression

$$\hat{\beta}^R = \arg\min_{\beta} \Big\{ \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij}\Big)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \Big\} = \arg\min_{\beta} \{\mathrm{RSS} + \lambda \|\beta\|_2^2\},$$

with tuning parameter $\lambda \geq 0$ (the intercept is not penalised). Ridge seeks coefficients that fit well (small RSS) but the *shrinkage penalty* is small only when $\beta_1, \dots, \beta_p$ are near zero. $\lambda = 0$ gives least squares; $\lambda \to \infty$ gives $\hat{\beta}^R \to 0$. λ is chosen by CV. For centred data $\hat{\beta}^R = (X^\top X + \lambda I)^{-1} X^\top Y$.

Scaling. Least squares is scale equivariant: multiplying X_j by c divides $\hat{\beta}_j$ by c, so $\hat{\beta}_j X_j$ is unchanged. Ridge estimates can change substantially, because the penalty treats all coefficients alike. Standardise first:

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}}.$$

On the Credit data, the standardised ridge coefficients shrink smoothly toward zero as λ grows (or as $\|\hat{\beta}^R_\lambda\|_2 / \|\hat{\beta}\|_2$ decreases from 1 to 0).

3.2 Why ridge improves over least squares

In a simulation with $n = 50$ and $p = 45$ (all coefficients nonzero), the squared bias increases and the variance decreases with λ, and the test MSE is minimised at an intermediate λ, well below that of least squares.

Theorem: bias and variance of ridge

For $Y = X\beta + \varepsilon$, $\varepsilon \sim (0, \sigma^2 I)$, and $\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top Y$,

$$\mathbb{E}\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top X\beta, \qquad \text{Bias} = -\lambda (X^\top X + \lambda I)^{-1}\beta,$$

$$\mathrm{Var}(\hat\beta_\lambda) = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1}, \qquad \text{MSE} = \|\text{Bias}\|^2 + \text{trVar}.$$

Moreover there is always some $\lambda > 0$ with $\mathrm{MSE}(\hat\beta_\lambda) < \mathrm{MSE}(\hat\beta_{\mathrm{OLS}})$ (Hoerl and Kennard, 1970).

Proof. The mean and variance follow from linearity in Y; $(X^\top X + \lambda I)^{-1} X^\top X - I = -\lambda (X^\top X + \lambda I)^{-1}$. With $X^\top X = V D^2 V^\top$ and $\theta = V^\top \beta$,

$$\mathrm{MSE}(\lambda) = \sum_{j=1}^{p} \frac{\sigma^2 d_j^2 + \lambda^2 \theta_j^2}{(d_j^2 + \lambda)^2}.$$

Differentiating, $\mathrm{MSE}'(\lambda) = \sum_j \frac{2 d_j^2 (\lambda \theta_j^2 - \sigma^2)}{(d_j^2 + \lambda)^3}$, which equals $-2\sigma^2 \sum_j d_j^{-4} < 0$ at $\lambda = 0$. So MSE decreases when moving from OLS ($\lambda = 0$) to small positive λ. ∎

As $\lambda \to \infty$ the bias grows and the variance vanishes; a proper λ balances the two, achieving lower MSE than OLS. The gain is largest when some d_j are small (collinearity) or p is close to n.

3.3 The lasso

Ridge keeps all p predictors in the model, which hampers interpretation when p is large. The lasso fixes this:

$$\hat\beta^L = \arg\min_\beta \{\mathrm{RSS} + \lambda \|\beta\|_1\}, \qquad \|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

Like ridge it shrinks toward zero, but the ℓ_1 penalty forces some coefficients to be **exactly zero** when λ is large enough. Like best subset, the lasso performs variable selection and gives **sparse** models; λ is chosen by CV. On the Credit data, coefficients enter the lasso path one at a time as λ decreases.

3.4 Constrained formulations and geometry

Ridge and lasso solve

$$\min_\beta \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_j \beta_j x_{ij}\Big)^2 \quad \text{subject to} \quad \sum_j \beta_j^2 \le s \quad \text{resp.} \quad \sum_j |\beta_j| \le s,$$

for some $s = s(\lambda)$ (Lagrange duality). Best subset is the same with $\sum_j \mathbb{I}(\beta_j \neq 0) \le s$. The RSS contours are ellipses around $\hat\beta_{\mathrm{OLS}}$; the lasso constraint is a diamond with corners on the axes, so the first contact often occurs at a corner where some $\beta_j = 0$; the ridge constraint is a disc with no corners, so contact is typically off the axes.

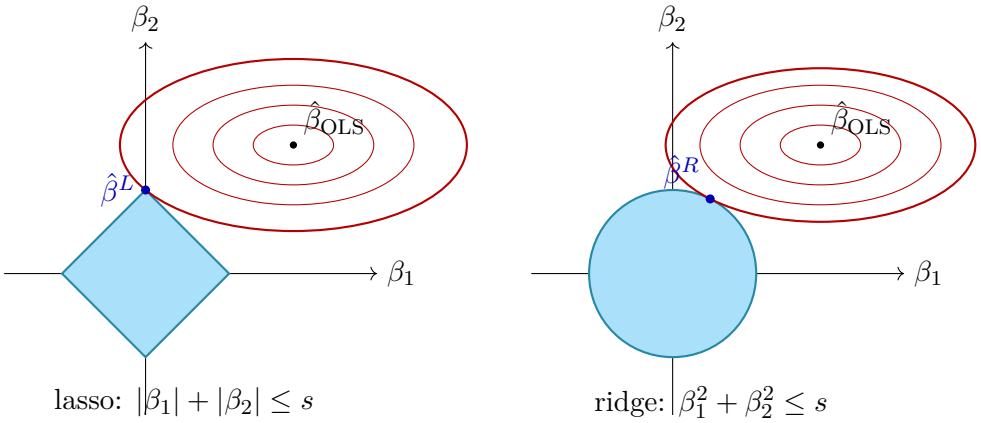


Figure 1: Contours of RSS (red ellipses) around $\hat{\beta}_{\text{OLS}}$ and the constraint regions (blue). The solution is the first point where an ellipse touches the region. The diamond's corners lie on the axes, so the lasso often lands on a corner ($\hat{\beta}_1 = 0$ here); the disc has no corners, so ridge shrinks both coefficients without zeroing either. (Contact points computed for the drawn ellipses.)

Theorem: orthonormal design

If $n = p$ and $X = I$ (no intercept), then for RSS $+ \lambda\cdot$penalty:

$$\hat{\beta}_j^R = \frac{y_j}{1+\lambda}, \qquad \hat{\beta}_j^L = \begin{cases} y_j - \lambda/2, & y_j > \lambda/2, \\ y_j + \lambda/2, & y_j < -\lambda/2, \\ 0, & |y_j| \le \lambda/2. \end{cases}$$

Proof. The objective separates: ridge minimises $(y_j - \beta_j)^2 + \lambda\beta_j^2$; lasso minimises $(y_j - \beta_j)^2 + \lambda|\beta_j|$, whose subgradient condition $-2(y_j - \beta_j) + \lambda s_j = 0$, $s_j \in \partial|\beta_j|$, gives the soft-thresholding rule. ■

Ridge shrinks every coefficient by the same proportion; the lasso subtracts a constant and truncates at zero (soft thresholding).

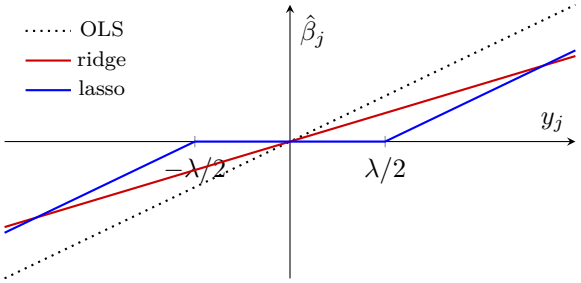


Figure 2: Orthonormal design: ridge rescales y_j by $1/(1+\lambda)$ (here $\lambda = 0.6$), the lasso soft-thresholds it, setting $|y_j| \le \lambda/2$ exactly to zero.

3.5 Dense versus sparse truth

When all 45 predictors matter (dense), ridge slightly beats the lasso; when only 2 of 45 matter (sparse), the lasso wins clearly. Neither universally dominates. Expect the lasso to do better when the response depends on relatively few predictors; but that number is never known a priori, so CV decides on a given data set.

3.6 Selecting the tuning parameter

Choose a grid of λ values, compute the CV error for each, pick the minimiser, and refit on all observations with the chosen λ.

4 Theoretical View in High Dimensions

Consider $Y = X\beta + \varepsilon$ with $X \in \mathbb{R}^{n\times p}$, possibly $p \gg n$. OLS is unstable or not identifiable; regularisation makes estimation possible by imposing structure: **ridge** assumes β dense but small in ℓ_2; the **lasso** assumes β sparse or approximately sparse. Their relative performance depends on the geometry of the true β.

Theorem: lasso rate under sparsity (informal)

Let β be s-sparse, $\|\beta\|_0 \le s$. Under a restricted-eigenvalue (RE) condition on X and with $\lambda \asymp \sigma\sqrt{\log p/n}$ (in the normalisation $\frac{1}{2n}\mathrm{RSS} + \lambda\|\beta\|_1$),

$$\|\hat\beta - \beta\|_2^2 \lesssim \frac{\sigma^2 s \log p}{n}$$

with high probability. Hence consistent estimation is possible when $n \gg s\log p$.

The price for not knowing the support is only a $\log p$ factor compared with the oracle rate $\sigma^2 s/n$, which explains why the lasso is effective for sparse signals.

Why design assumptions are needed. Sparsity alone is not sufficient: if columns of X are highly correlated, different sparse vectors produce similar fitted values. Guarantees require conditions such as restricted eigenvalue, compatibility, or (stronger) restricted isometry; exact support recovery needs even stronger assumptions — an irrepresentable condition and a beta-min condition ($\min_{j\in S}|\beta_j|$ above the noise level).

Ridge, $\hat\beta^{\text{ridge}} = \arg\min\{\|Y - X\beta\|_2^2 + \lambda\|\beta\|_2^2\}$, does not exploit sparsity but stabilises estimation under multicollinearity; it is favourable when β is dense with small ℓ_2 norm, where shrinking all coefficients slightly beats enforcing sparsity.

Effective complexity. For sparse vectors $B_0(s) = \{\beta : \|\beta\|_0 \le s\}$ the complexity is essentially $s\log p$; for dense vectors $B_2(R) = \{\beta : \|\beta\|_2 \le R\}$ it is governed by ℓ_2 geometry. The lasso is adapted to ℓ_1-type sparsity, ridge to ℓ_2-type regularity.

5 Dimension Reduction Methods

5.1 General idea

Construct $M < p$ linear combinations

$$Z_m = \sum_{j=1}^{p} \phi_{mj} X_j, \qquad m = 1, \dots, M,$$

and fit $Y = \theta_0 + \sum_{m=1}^{M} \theta_m Z_m + \varepsilon$ by least squares. Each Z_m is a direction in feature space; the choice of ϕ_{mj} determines the method. Since

$$\sum_{m=1}^{M} \theta_m Z_m = \sum_{j=1}^{p} \beta_j X_j, \qquad \beta_j = \sum_{m=1}^{M} \theta_m \phi_{mj},$$

dimension reduction constrains β to an M-dimensional subspace, which reduces variance (at the cost of bias if β lies outside it).

5.2 Principal components regression

PCA defines the directions: the first principal component is the normalised linear combination with the largest variance; the second has the largest variance subject to being uncorrelated with the first, and so on (eigenvectors of the sample covariance matrix). With many correlated variables we replace them by a few components capturing their joint variation.

Example: advertising data, population and ad spending in 100 cities

$Z_1 = 0.839(\texttt{pop} - \overline{\texttt{pop}}) + 0.544(\texttt{ad} - \overline{\texttt{ad}})$, with $0.839^2 + 0.544^2 = 1$. The first PC is the line that minimises the sum of squared perpendicular distances to the points; its scores z_{i1} are strongly related to both pop and ad, while the second-PC scores are only weakly related.

Assumption: the directions in which $X_1, \ldots, X_p$ vary most are the directions associated with Y. When the response depends on the first few PCs, PCR does well; when it depends on many predictors in a sparse way, PCR needs many components. PCR with $M = p$ is least squares; PCR is close in spirit to ridge (ridge shrinks low-variance directions smoothly, PCR drops them). On the Credit data, the CV error is lowest near $M = 10$, i.e. little reduction.

Standardisation: generally standardise before PCA; otherwise high-variance variables dominate the components and the measurement scale affects the final PCR model. If all variables share the same unit one may skip it.

Limitation: the directions are found *unsupervised* — the response does not supervise them — so the directions that best explain X need not be best for predicting Y.

5.3 Partial least squares

PLS also builds $Z_1, \ldots, Z_M$ as linear combinations and fits least squares, but chooses them *supervised*: directions that approximate the predictors *and* relate to the response.

Theorem: the first PLS direction

With standardised X and centred Y,

$$w_1 = \arg\max_{\|w\|=1} \operatorname{Cov}(Xw, Y)^2 \;\propto\; X^\top Y,$$

i.e. ϕ_{1j} is proportional to the simple-regression coefficient of Y on X_j; $Z_1 = Xw_1$.

Proof. $\operatorname{Cov}(Xw, Y) = \frac{1}{n} w^\top X^\top Y$, and by Cauchy–Schwarz $(w^\top X^\top Y)^2 \le \|w\|^2 \|X^\top Y\|^2$ with equality iff $w \propto X^\top Y$. For standardised X_j the simple-regression slope of Y on X_j is $x_{(j)}^\top Y / n$. ■

Algorithm: PLS (sequential construction)

Standardise X and Y; set $X^{(0)} = X$, $Y^{(0)} = Y$. For $m = 1, \ldots, M$:

1. direction $w_m \propto X^{(m-1)\top} Y^{(m-1)}$;
2. component $Z_m = X^{(m-1)} w_m$;
3. regress and update residuals $Y^{(m)} = Y^{(m-1)} - \hat{\theta}_m Z_m$;

4. deflate the predictors $X^{(m)} = X^{(m-1)} - Z_m p_m^\top$, $p_m = X^{(m-1)\top} Z_m / (Z_m^\top Z_m)$ (each column is replaced by its residual after regressing on Z_m).

So $Y = \theta_1 Z_1 + E_1$, then $E_1 = \theta_2 Z_2 + E_2$, and so on. The components are mutually uncorrelated. On the advertising data, the first PLS direction puts more weight on `ad` than the first PC, since `ad` is more related to sales. In practice PLS often performs no better than ridge or PCR: supervision reduces bias but can increase variance.

5.4 Theoretical comparison

Method	Assumption on β	Mechanism
lasso	sparsity	ℓ_1 constraint
ridge	small ℓ_2 norm	shrinkage
PLS	low-dimensional structure	iterative projection

PLS does not assume sparsity but that β lies in a low-dimensional subspace. With $\Sigma = \mathrm{Cov}(X)$ and $c = \mathrm{Cov}(X, Y)$, the m-component PLS estimator lies in the Krylov subspace

$$\mathcal{K}_m(\Sigma, c) = \mathrm{span}\{c, \Sigma c, \ldots, \Sigma^{m-1} c\},$$

(it equals least squares restricted to that subspace, i.e. m steps of conjugate gradients on the normal equations), and it is consistent if $\beta \in \mathcal{K}_m(\Sigma, c)$, i.e. when the signal has low-dimensional structure. Since $\beta = \Sigma^{-1} c$, this holds e.g. when c lies in the span of m eigenvectors of Σ.

Summary

- Best subset is accurate but expensive; forward/backward stepwise search $1+p(p+1)/2$ models; the final size is chosen by C_p, AIC, BIC, adjusted R^2 or CV.
- Ridge shrinks all coefficients (always beats OLS for some λ); the lasso shrinks and selects; standardise first and choose λ by CV.
- In high dimensions the lasso attains $s \log p / n$ under design conditions; ridge suits dense signals.
- PCR regresses on unsupervised principal components; PLS uses supervised directions maximising covariance with Y (Krylov subspace).

Chapter 7. Nonlinear Regression

Lecture 6-4 (polynomials, step functions, basis functions, regression splines, smoothing splines, local regression, GAM)

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1 Basis-Function Regression

1.1 Polynomial regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i.$$

For large enough d this produces an extremely nonlinear curve, yet the parameters are estimated by OLS: create $X_1 = X$, $X_2 = X^2$, ... and treat it as multiple linear regression. We are rarely interested in the coefficients; we care about the fitted function

$$\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4,$$

with pointwise interval $\hat{f}(x_0) \pm t_{1-\alpha/2,\mathrm{df}}\,\mathrm{SE}[\hat{f}(x_0)]$, where $\mathrm{SE}[\hat{f}(x_0)]^2 = \ell_0^\top \widehat{\mathrm{Var}}(\hat{\beta})\ell_0$ with $\ell_0 = (1, x_0, \ldots, x_0^4)^\top$. Fix d at a low value (rarely above 3 or 4) or choose it by CV. Example: a degree-4 polynomial of wage on age, with 95% bands; on the Auto data, $d = 1$ underfits mpg on horsepower, $d = 2$ fits well, $d = 14$ is wild.

Limitation. Polynomials are global: the fit in one region is affected by data far away, the tails become erratic for large d, and a polynomial may fail to recover a nonlinear pattern even when d is huge.

1.2 Step functions

Cut the range of X at $c_1 < \cdots < c_K$:

$$C_0(X) = \mathbb{I}(X < c_1),\ C_1(X) = \mathbb{I}(c_1 \le X < c_2),\ \ldots,\ C_K(X) = \mathbb{I}(c_K \le X),$$

so $C_0 + \cdots + C_K = 1$, and fit $y_i = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$ (C_0 is dropped because of the intercept). β_0 is the mean for $X < c_1$ and β_k the difference of bin k from it. With bins < 35, 35–50, 50–65, ≥ 65 for wage on age, the first bin misses the increasing trend; still, step functions are popular in biostatistics and epidemiology (e.g. five-year age groups). The result depends on the breakpoints.

1.3 Basis functions

Both are special cases of

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

with fixed, known functions b_k: $b_j(x) = x^j$ for polynomials and $b_j(x) = \mathbb{I}(c_j \le x < c_{j+1})$ for step functions. Everything of linear regression (least squares, SEs, F-tests) applies. Other bases: Fourier series, wavelets (e.g. for image denoising), and the splines below.

2 Regression Splines

2.1 Piecewise polynomials

Fit separate low-degree polynomials over regions separated by **knots**; with one knot c,

$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \varepsilon_i, & x_i < c, \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \varepsilon_i, & x_i \ge c. \end{cases}$$

More knots give more flexibility. Unconstrained pieces jump at the knot, so we add constraints (continuity, continuity of derivatives). Splines have the "maximum" amount of continuity.

2.2 Linear splines

Definition: linear spline

A linear spline with knots $\xi_1 < \cdots < \xi_K$ is piecewise linear and continuous at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+1} b_{K+1}(x_i) + \varepsilon_i, \qquad b_1(x) = x, \quad b_{k+1}(x) = (x - \xi_k)_+,$$

with $(x - \xi)_+ = x - \xi$ if $x > \xi$ and 0 otherwise.

Each truncated term $(x - \xi_k)_+$ adds a change of slope β_{k+1} at ξ_k while keeping continuity.

(a) basis functions (b) linear spline



Figure 1: (a) The truncated power basis of a linear spline with knots ξ_1, ξ_2. (b) A fit $\beta_0 + \beta_1 x + \beta_2(x - \xi_1)_+ + \beta_3(x - \xi_2)_+$: continuous, with slope changes β_2 at ξ_1 and β_3 at ξ_2.

2.3 Cubic splines

Definition: cubic spline and truncated power basis

A cubic spline with knots $\xi_1, \ldots, \xi_K$ is piecewise cubic with continuous derivatives up to order 2 at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i,$$

$$b_1(x) = x, \; b_2(x) = x^2, \; b_3(x) = x^3, \; b_{k+3}(x) = (x - \xi_k)_+^3, \; k = 1, \ldots, K.$$

Theorem: degrees of freedom of a cubic spline

The space of cubic splines with K knots has dimension $K + 4$, and the truncated power basis (with the constant) is a basis.

Proof. $K + 1$ unconstrained cubic pieces have $4(K + 1)$ parameters; continuity of the function, first and second derivatives at each of the K knots imposes $3K$ linear constraints, leaving $4(K+1) - 3K = K + 4$. Each $(x - \xi_k)_+^3$ is twice continuously differentiable (its first two derivatives vanish at ξ_k) and its third derivative jumps by 6 at ξ_k, so the $K + 4$ functions are linearly independent splines. ■

With continuity alone the fit can have a kink; continuity of the first and second derivatives makes it visually smooth.

2.4 Natural cubic splines

Splines can have high variance at the outer range of the predictor (few data, wide confidence bands). A **natural cubic spline** additionally requires the function to be *linear* beyond the boundary knots.

This adds 4 constraints (second and third derivatives zero at both ends), so a natural cubic spline with K knots has K degrees of freedom (including the intercept), and its confidence bands are narrower at the boundaries.

2.5 Number and locations of knots

Decide the number K (or the degrees of freedom) and place knots at quantiles of the observed X (or let software do it), so that each region has a similar amount of data. Choose K by cross-validation. On the Wage data, a degree-15 polynomial is wild at the boundaries, whereas a natural cubic spline with 15 df is well behaved: splines add flexibility through knots while keeping the degree fixed.

3 Smoothing Splines

Definition: smoothing spline

Among all twice-differentiable functions g,

$$\min_g \underbrace{\sum_{i=1}^{n}(y_i - g(x_i))^2}_{\text{RSS}} + \lambda \underbrace{\int g''(t)^2\,dt}_{\text{roughness penalty}} .$$

The first term makes g match the data; the second controls how wiggly g is (g'' measures curvature), modulated by $\lambda \geq 0$. The smaller λ, the more wiggly the function, eventually interpolating the y_i at $\lambda = 0$; as $\lambda \to \infty$, g becomes linear (any linear function has zero penalty). λ can be chosen by K-fold CV.

Theorem: the minimiser is a natural cubic spline

The minimiser $\hat{g}_\lambda$ is a natural cubic spline with knots at the unique values of $x_1, \ldots, x_n$ — a shrunken version of the interpolating natural spline. Consequently $\hat{g}_\lambda = (\hat{g}(x_1), \ldots, \hat{g}(x_n))^\top = S_\lambda y$ for an $n \times n$ smoother matrix S_λ.

Idea. Given any candidate g, let $\tilde{g}$ be the natural cubic spline interpolating the values $g(x_i)$. Then RSS is unchanged, and an integration-by-parts argument shows $\int \tilde{g}''^2 \leq \int g''^2$, with equality only if $g = \tilde{g}$. So the search can be restricted to natural cubic splines with knots at the x_i, which form an n-dimensional linear space $g(x) = \sum_j N_j(x)\theta_j$. The problem becomes $\min_\theta \|y - N\theta\|^2 + \lambda\theta^\top \Omega_N \theta$ with $(\Omega_N)_{jk} = \int N_j'' N_k''$, solved by $\hat{\theta} = (N^\top N + \lambda\Omega_N)^{-1} N^\top y$, a generalised ridge regression. Hence $S_\lambda = N(N^\top N + \lambda\Omega_N)^{-1} N^\top$. ■

The **effective degrees of freedom** $\mathrm{df}_\lambda = \mathrm{tr}(S_\lambda)$ decreases from n ($\lambda = 0$) to 2 ($\lambda \to \infty$) and is a convenient way to specify λ. Because $\hat{g}$ is a linear smoother, LOOCV costs one fit: $\mathrm{CV}(\lambda) = \frac{1}{n}\sum_i \{(y_i - \hat{g}_\lambda(x_i))/(1 - \{S_\lambda\}_{ii})\}^2$ (Ch. 3). On the Auto and Wage data, a smoothing spline with about 6.8 effective df (chosen by LOOCV) is nearly indistinguishable from a 16-df fit.

4 Local Regression

With a sliding weight function, fit separate (weighted) linear fits over the range of X.

Algorithm: local regression at a target point x_0

1. Gather the fraction $s = k/n$ of training points closest to x_0 (the **span**).
2. Assign weights $K_{i0} = K(x_i, x_0)$: zero outside the neighbourhood, largest for the closest points (e.g. tricube weights).
3. Fit a weighted least-squares regression $\min_{\beta_0,\beta_1} \sum_i K_{i0}(y_i - \beta_0 - \beta_1 x_i)^2$.
4. The fitted value is $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Choices: the weighting function, whether to fit a constant, linear or quadratic regression locally, and the span s, which plays the role of λ: small s gives a wiggly fit, large s a global one; choose s by CV. Local linear fits remove the boundary bias that local constant (kernel) averages have. Like KNN, local regression suffers from the curse of dimensionality for more than three or four predictors.

5 Generalised Additive Models

GAMs allow flexible nonlinearities in several variables while retaining the additive structure of linear models:

$$y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \varepsilon_i.$$

A GAM can be fitted with natural splines (then it is one big least-squares regression on the combined basis), with smoothing splines or local regression, or mixing terms — some linear, some nonlinear, some step functions.

Algorithm: backfitting (for smoothing-spline or local-regression terms)

Initialise $\hat{\beta}_0 = \bar{y}$, $\hat{f}_j = 0$. Repeat until convergence: for $j = 1, \ldots, p$, smooth the partial residuals $r_i = y_i - \hat{\beta}_0 - \sum_{k \neq j} \hat{f}_k(x_{ik})$ against x_{ij} to update $\hat{f}_j$, then centre $\hat{f}_j$ to have mean zero.

Each f_j is updated with the others held fixed, which is possible because the model is additive.

Example: Wage data

$\texttt{wage} = \beta_0 + f_1(\texttt{year}) + f_2(\texttt{age}) + f_3(\texttt{education}) + \varepsilon$, with natural splines for year and age and a separate constant for each education level. Holding the others fixed, wage rises slightly with year, rises and then falls with age, and increases with education.

Pros. A nonlinear f_j for each X_j without manually trying many transformations; the additive structure still allows examining the effect of each X_j with the others fixed; potentially more accurate predictions. Effective degrees of freedom summarise each f_j. **Cons.** Additivity can miss important interactions (they can be added by hand, e.g. $f_{jk}(X_j, X_k)$ via a two-dimensional smoother).

Summary

- Basis-function regression (polynomials, step functions, splines) stays within linear least squares.
- Splines are piecewise polynomials joined smoothly at knots: linear splines $(x - \xi)_+$, cubic splines $(x - \xi)_+^3$ with $K + 4$ df, natural cubic splines linear beyond the boundary (K df).

- Smoothing splines $\min \mathrm{RSS} + \lambda \int g''^2$ yield natural cubic splines with knots at the data; $\mathrm{df}_\lambda = \mathrm{tr} S_\lambda$.
- Local regression fits weighted least squares in a sliding window (span s).
- GAMs combine univariate nonlinear functions additively (backfitting); interactions must be added explicitly.

Chapter 8. Classification Errors and the Bayes Classifier

Lecture 7-1 (error rate, Bayes optimal classifier, KNN, bias–variance, confusion matrix, ROC/AUC, CV for classification)

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1 The Classification Problem

For regression we assessed accuracy by MSE; for classification we use the **error rate**

$$\text{Error rate} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(y_i \neq \hat{y}_i),$$

the proportion of misclassifications. Goals: build a classifier $C(X)$ that assigns a class label from $\mathcal{C}$ to a future unlabelled observation; assess the uncertainty of each classification; understand the roles of the predictors.

2 The Bayes Optimal Classifier

Let the K classes be numbered $1, \ldots, K$. We want a classifier with small misclassification risk $R(C) = \Pr(C(X) \neq Y)$.

Theorem: Bayes classifier

The risk is minimised by

$$C^*(x) = \arg\max_k \Pr(Y = k \mid X = x),$$

assigning each x to the class with the largest conditional probability. Its risk $R(C^*) = 1 - \mathbb{E}\big[\max_k \Pr(Y = k \mid X)\big]$ is the **Bayes error rate**, the lowest error achievable by any classifier.

Proof. $R(C) = 1 - \Pr(C(X) = Y) = 1 - \mathbb{E}\big[\Pr(Y = C(X) \mid X)\big]$. For each x, $\Pr(Y = C(x) \mid X = x) \leq \max_k \Pr(Y = k \mid X = x)$, with equality when $C(x) = C^*(x)$. Taking expectations, $R(C) \geq R(C^*)$. ■

The Bayes error is what we would get if we knew the true distribution; it is positive whenever the classes overlap. In a simulated two-class example (100 observations per class), the **Bayes decision boundary** $\{x : \Pr(Y = 1 \mid x) = 0.5\}$ is a curved line.

3 Approximating the Bayes Classifier

In finite samples we measure a classifier $\hat{C}$ by its test error,

$$\text{Err}_{\text{Te}} = \text{Ave}_{i \in \text{Te}}\, \mathbb{I}\{y_i \neq \hat{C}(x_i)\}.$$

The Bayes classifier has the smallest population error; a good classifier has small test error. Two strategies: *estimate the conditional probabilities* and plug them into the Bayes rule (logistic regression, GAMs, LDA/QDA, KNN), or *build structured models for $C(x)$ directly* (support-vector machines).

3.1 K-nearest neighbours

KNN is a flexible approach to estimate the Bayes classifier. For a new point x, look at its neighbours:

$$\hat{\Pr}(Y = k \mid X = x) = \frac{1}{K}\sum_{x_i \in \mathcal{N}_K(x)} \mathbb{I}(y_i = k),$$

and predict the majority class among the K closest training points (for two classes coded 0/1, $\hat{f}(x) = \frac{1}{K}\sum y_i$ and classify by $\hat{f} > 0.5$). K controls complexity: **the smaller K, the more complex** the model. With $K = 1$ the boundary is overly flexible (training error 0); with $K = 100$ it is nearly linear and not flexible enough; $K = 10$ is close to the Bayes boundary.

3.2 Bias–variance in classification

As complexity increases ($1/K$ grows), training error always declines, while test error first declines (bias reduction dominates) and then increases (variance dominates): a U-shape. More flexible is not always better.

Remark: a classic result

As $n \to \infty$, the error of the 1-nearest-neighbour rule is at most twice the Bayes error (Cover and Hart, 1967): for two classes, $R_{1\text{NN}} \to \mathbb{E}[2\eta(X)(1-\eta(X))] \leq 2R^*(1-R^*)$ with $\eta(x) = \Pr(Y = 1 \mid x)$.

4 Other Measures for Classification

4.1 Confusion matrix

Default data (LDA fit, $n = 10{,}000$):

		True default status		
		No	Yes	Total
Predicted	No	9,644	252	9,896
	Yes	23	81	104
	Total	9,667	333	10,000

- Overall error: $(23 + 252)/10000 = 2.75\%$ — but always predicting "No" would already give $333/10000 = 3.33\%$.
- **False positive rate**: fraction of negatives classified positive, $23/9667 = 0.2\%$.
- **False negative rate**: fraction of positives classified negative, $252/333 = 75.7\%$.

Sensitivity (recall, true positive rate) $= 1-\text{FNR} = 81/333 = 24.3\%$; specificity $= 1-\text{FPR} = 99.8\%$. A bank that wants to catch defaulters finds this unacceptable even though the overall error is low.

4.2 Thresholds

Classifying "Yes" when $\hat{\Pr}(\text{Yes} \mid x) \geq 0.5$ mimics the Bayes rule for overall error. Lowering the threshold (e.g. to 0.2 or 0.1) reduces the false negative rate at the cost of more false positives and possibly a higher overall error. The choice depends on the costs of the two errors: with costs c_{FN} and c_{FP}, predict "Yes" iff $\hat{p}\, c_{FN} > (1-\hat{p})c_{FP}$, i.e. $\hat{p} > c_{FP}/(c_{FP} + c_{FN})$.

4.3 ROC curve and AUC

The **ROC curve** plots the true positive rate against the false positive rate as the threshold varies from 1 to 0, displaying both error types simultaneously. The **AUC** (area under the curve) summarises overall performance; higher is better, 0.5 is random guessing and 1 is perfect separation.

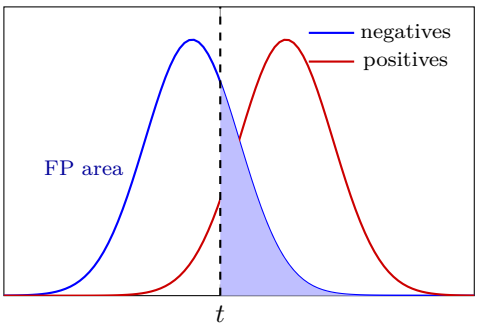
Theorem: probabilistic meaning of AUC

For a score $s(x)$, $\text{AUC} = \Pr\left(s(X^+) > s(X^-)\right)$ for independent draws X^+ from the positive and X^- from the negative class (ties counted 1/2).

Proof. With threshold t, $\text{TPR}(t) = \Pr(s^+ > t)$ and $\text{FPR}(t) = \Pr(s^- > t)$. Then $\text{AUC} = \int \text{TPR}\, d\text{FPR} = \int \Pr(s^+ > t)\, dF_-(t) = \Pr(s^+ > s^-)$. ■

AUC depends only on the ranking of scores, so it is invariant to monotone transformations and to the class proportions.

(a) score distributions



(b) ROC curve

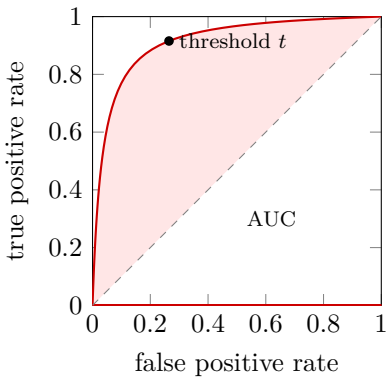


Figure 1: (a) Scores of negatives and positives; classifying "positive" above a threshold t gives a false positive rate (shaded) and a true positive rate. (b) Moving t from $+\infty$ to $-\infty$ traces the ROC curve; the area below it is the AUC. The diagonal is random guessing (AUC = 0.5).

5 Cross-Validation for Classification

Divide the data into K roughly equal parts $C_1, \ldots, C_K$ with n_k observations in part k and compute

$$\text{CV}_K = \sum_{k=1}^{K} \frac{n_k}{n} \text{Err}_k, \qquad \text{Err}_k = \frac{1}{n_k} \sum_{i \in C_k} \mathbb{I}(y_i \neq \hat{y}_i),$$

with $\hat{y}_i$ from the model fitted without part k. On two-dimensional simulated data, for logistic regression with polynomial predictors of increasing order and for KNN with decreasing K, the 10-fold CV curve tracks the true test error closely (it may underestimate the level, but its minimum is near the right complexity), while the training error keeps falling.

Summary

- The Bayes classifier $\arg\max_k \Pr(Y = k \mid x)$ minimises the error rate; its error is a lower bound.
- KNN estimates the conditional probabilities by local voting; small K is flexible, large K smooth.
- The overall error rate can hide poor performance on a rare class; report FPR, FNR, sensitivity and specificity, and choose thresholds by costs.
- ROC curves show all thresholds; AUC is the probability of ranking a random positive above a random negative.

- CV with the misclassification loss selects complexity for classifiers.

Chapter 9. Discriminative Methods: Logistic Regression

Lecture 7-2 (discriminative vs generative, logit model, GLM, maximum likelihood, confounding, extensions, multinomial regression)

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1 Discriminative Methods versus Generative Models

Discriminative: model $\Pr(Y = c_k \mid X = x)$ directly	Generative: model (X, Y), then derive $\Pr(Y = c_k \mid X = x)$
K-nearest neighbours, logistic regression, classification trees, ensembles (bagging, random forest, boosting), support vector machines	linear and quadratic discriminant analysis, naive Bayes, Gaussian mixture models

Examples of classification: tumour cells benign/malignant; credit-card transactions legitimate/fraudulent; news stories finance/weather/entertainment/sports. We observe (X, Y) with Y a class label and learn $C(X)$ that predicts Y; often we estimate $P(Y = k \mid X)$ and classify to the most probable class.

2 Why Not Linear Regression?

Code $Y = 0$ (No) and $Y = 1$ (Yes) and fit $\hat{Y} = \beta_0 + \beta X$, classifying by $\hat{Y} > 0.5$? There is a model mismatch: $\mathbb{E}[Y \mid X] = P(Y = 1 \mid X)$ must lie in $[0, 1]$, but a linear model is unbounded and can give invalid probabilities; the relationship is typically nonlinear in X. (For two classes, the least-squares fit is in fact equivalent to LDA for the classification rule, but the probability estimates are poor.)

With more than two classes, coding e.g. $Y = 1$ (stroke), 2 (drug overdose), 3 (epileptic seizure) imposes an ordering and equal spacing that has no meaning, and different codings give different fits. Multiclass logistic regression or discriminant analysis is appropriate.

3 The Logistic Model

For two classes $Y \in \{c_1, c_2\}$, model $\Pr(Y = c_1 \mid X)$ through the **logit** transformation:

$$\log \frac{\Pr(Y = c_1 \mid X)}{\Pr(Y = c_2 \mid X)} = \beta_0 + \beta_1 X,$$

$$\Pr(Y = c_1 \mid X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}, \quad \Pr(Y = c_2 \mid X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X}},$$

which keeps probabilities in $[0, 1]$ and gives an S-shaped curve. β_1 is the change in the **log odds** for a one-unit increase in X; e^{β_1} multiplies the odds; the change in the probability itself depends on X, $\partial p / \partial X = \beta_1 p(1 - p)$.

3.1 Logistic regression as a GLM

A generalised linear model consists of a random component (Y from an exponential family), a linear predictor $\eta = X^\top \beta$, and a link $g(\mathbb{E}[Y]) = \eta$. Logistic regression has $Y \sim \text{Bernoulli}(p)$ and the logit link $g(p) = \log \frac{p}{1-p}$, which is the canonical link of the Bernoulli family (the natural parameter of $p^y (1 - p)^{1-y} = \exp\{y \log \frac{p}{1-p} + \log(1 - p)\}$). Hence $\log \frac{p}{1-p} = X^\top \beta$.

4 Estimation by Maximum Likelihood

The likelihood of the observed zeros and ones is

$$\mathcal{L}(\beta_0, \beta) = \prod_{i: y_i = 1} p(x_i) \prod_{j: y_j = 0} \{1 - p(x_j)\}, \qquad \ell(\beta_0, \beta_1) = \sum_i \Big[y_i (\beta_0 + \beta_1 x_i) - \log \left(1 + e^{\beta_0 + \beta_1 x_i}\right) \Big].$$

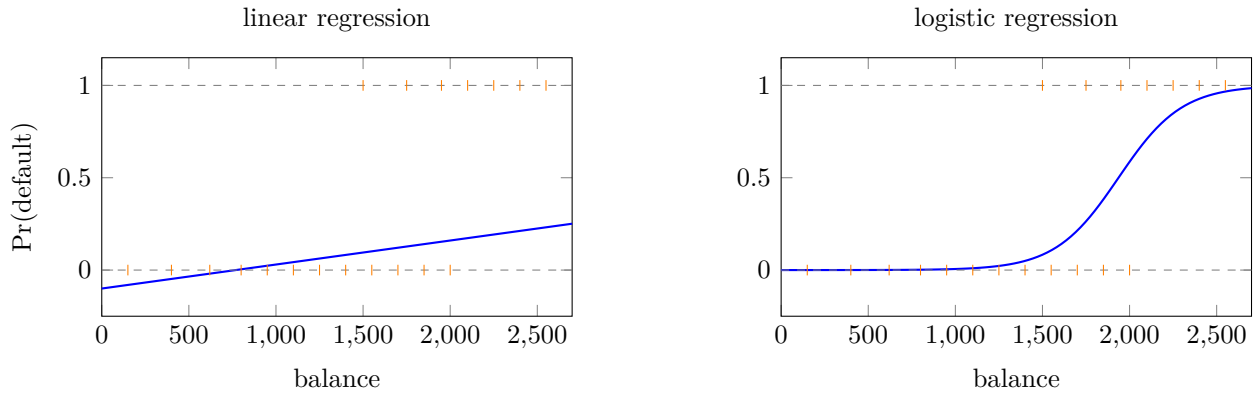


Figure 1: Default data (schematic points; orange ticks are $Y = 0$ at the bottom and $Y = 1$ at the top). A linear fit can give probabilities below 0 (and above 1); the fitted logistic curve $\hat{p} = e^{-10.65+0.0055X}/(1 + e^{-10.65+0.0055X})$ stays in $[0, 1]$.

Theorem: score equations and concavity

With $p_i = p(x_i)$ and design rows $\tilde{x}_i = (1, x_i)^\top$,

$$\nabla \ell = \sum_i (y_i - p_i)\tilde{x}_i = \tilde{X}^\top (y - p), \qquad \nabla^2 \ell = -\tilde{X}^\top W \tilde{X}, \; W = \mathrm{diag}\{p_i(1 - p_i)\}.$$

ℓ is concave, so the MLE solves $\tilde{X}^\top(y-p) = 0$ and is found by Newton–Raphson, equivalently **iteratively reweighted least squares**: $\beta^{\text{new}} = (\tilde{X}^\top W \tilde{X})^{-1} \tilde{X}^\top W z$ with working response $z = \tilde{X}\beta + W^{-1}(y - p)$.

Proof. $\frac{d}{d\eta} \log(1 + e^\eta) = \frac{e^\eta}{1+e^\eta} = p$ and $\frac{dp}{d\eta} = p(1-p)$; differentiate termwise. The Hessian is negative semidefinite. A Newton step $\beta + (\tilde{X}^\top W \tilde{X})^{-1} \tilde{X}^\top (y - p)$ rearranges into the weighted least-squares form. ■

The asymptotic covariance of $\hat{\beta}$ is $(\tilde{X}^\top \hat{W} \tilde{X})^{-1}$ (inverse Fisher information), which gives the standard errors and z-statistics. If the classes are perfectly separable, the likelihood increases without bound as $\|\beta\| \to \infty$ and the MLE does not exist (estimates become unstable — see also Ch. 10).

5 Logistic Regression on the Default Data

	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.6513	0.3612	−29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

A one-unit increase in balance increases the log odds of default by 0.0055. The z-statistic tests H_0: no relationship between X and the odds, i.e. $H_0 : \beta_1 = 0$; here $p < 0.0001$.

Predictions.

$$\hat{p}(1000) = \frac{e^{-10.6513+0.0055\times 1000}}{1 + e^{-10.6513+0.0055\times 1000}} = 0.006, \qquad \hat{p}(2000) = \frac{e^{-10.6513+0.0055\times 2000}}{1 + e^{-10.6513+0.0055\times 2000}} = 0.586.$$

A qualitative predictor. With student status: intercept −3.5041, student[Yes] 0.4049 ($p = 0.0004$), so $\Pr(\text{default} \mid \text{student}) = e^{-3.0992}/(1 + e^{-3.0992}) = 0.0431$ and $\Pr(\text{default} \mid \text{non-student}) = e^{-3.5041}/(1 + e^{-3.5041}) = 0.0292$.

6 Multiple Logistic Regression and Confounding

$$\log \frac{p(X)}{1-p(X)} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p, \qquad p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}.$$

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income (thousands)	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

Why is the student coefficient now *negative* when it was positive alone? **Confounding**: students tend to carry higher balances, so their marginal default rate is higher; but at each fixed balance, students default *less* than non-students. Multiple logistic regression teases the two apart: the marginal association mixes the direct (negative) effect with the indirect path through balance.

7 Extensions of Logistic Regression

- **Polynomial logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d$. Modelling the event $\texttt{wage} > 250$ with a degree-4 polynomial in age gives the fitted probability with 95% bands (wide where data are scarce).
- **Piecewise-constant logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i)$ with step functions of age.
- **GAM for classification**: $\log \frac{P(X)}{1-P(X)} = \beta_0 + f_1(X_1) + \cdots + f_p(X_p)$; for the Wage data, $\beta_1\,\texttt{year} + f_2(\texttt{age}) + f_3(\texttt{education})$ with a smoothing spline (5 df) in age and a step function for education.
- **Penalised logistic regression**: minimise $-\log \mathcal{L}(\beta_0, \beta) + \lambda\|\beta\|_2^2$ (ridge) or $+\lambda\|\beta\|_1$ (lasso), generally $+\lambda\|\beta\|_q^q$; the penalty also gives finite estimates under separation.
- **Multinomial logistic regression** (below).

8 More Than Two Classes: Multinomial Regression

With K classes (e.g. stroke, drug overdose, epileptic seizure), choose class K as baseline and model, for $k = 1, \ldots, K-1$,

$$\log \frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)} = \beta_{k0} + \beta_{k1} x_1 + \cdots + \beta_{kp} x_p.$$

Theorem: implied probabilities

$$\Pr(Y = k \mid X = x) = \frac{e^{\beta_{k0}+\beta_k^\top x}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_l^\top x}} \quad (k < K),$$

$$\Pr(Y = K \mid X = x) = \frac{1}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_l^\top x}},$$

and the log odds between any two classes k, k' is linear: $\log \frac{\Pr(Y=k|x)}{\Pr(Y=k'|x)} = (\beta_{k0} - \beta_{k'0}) + (\beta_k - \beta_{k'})^\top x$.

Proof. Write $\pi_k = \Pr(Y = k \mid x) = \pi_K e^{\eta_k}$ for $k < K$; the probabilities sum to 1, so $\pi_K(1+\sum_l e^{\eta_l}) = 1$. The difference of two log odds against K gives the pairwise log odds. ■

The choice of baseline does not affect the fitted probabilities, only the parametrisation. A symmetric version (softmax) uses $\Pr(Y = k \mid x) = e^{\eta_k} / \sum_{l=1}^{K} e^{\eta_l}$ with one set of coefficients per class (identifiable only up to a common shift).

Summary

- Discriminative methods model $\Pr(Y \mid X)$ directly; generative methods model $\Pr(X \mid Y)\Pr(Y)$.
- Logistic regression models the log odds linearly (a Bernoulli GLM with logit link) and is fitted by maximum likelihood (IRLS).
- Coefficients are changes in log odds; multiple regression reveals confounding (student effect reverses sign given balance).
- Polynomial, step-function, GAM and penalised versions extend it; multinomial regression handles $K > 2$ classes with linear pairwise log odds.

Chapter 10. Generative Models for Classification

Lecture 7-3 (Bayes theorem, LDA, QDA, naive Bayes, logistic regression vs LDA, comparison of classifiers)

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1 Discriminant Analysis as Generative Modelling

Generative models describe the joint distribution

$$\Pr(X, Y) = \Pr(X \mid Y)\Pr(Y)$$

(e.g. Gaussian mixture models) and classify by Bayes' rule $\Pr(Y \mid X) = \Pr(X \mid Y)\Pr(Y)/\Pr(X)$. Discriminant analysis is the special case that assumes a specific form for $\Pr(X \mid Y)$ (e.g. Gaussian) and uses it to derive $\Pr(Y \mid X)$: (1) model the distribution of X in each class separately, (2) flip with Bayes' theorem. Gaussian classes give

$$X \mid Y = c_k \sim N(\mu_k, \Sigma) \;\; \text{(LDA)}, \qquad X \mid Y = c_k \sim N(\mu_k, \Sigma_k) \;\; \text{(QDA)};$$

other distributions can be used as well.

1.1 Bayes' theorem for classification

Let $f_k(x)$ be the density of X in class k and $\pi_k = \Pr(Y = k)$ the prior probability. Then

$$p_k(x) = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)} \;\propto\; \pi_k f_k(x).$$

Classify to the highest posterior: $\hat{C}(x) = \arg\max_k \pi_k f_k(x)$ (the Bayes classifier of Ch. 8 with estimated ingredients). Unequal priors shift the boundary: favouring one class moves the boundary away from it.

1.2 Why discriminant analysis?

When the classes are well separated, logistic-regression estimates are surprisingly unstable (they diverge under perfect separation); LDA does not suffer from this. If n is small and X is approximately normal in each class, LDA is again more stable. LDA is popular with more than two classes and provides low-dimensional views of the data.

2 Linear Discriminant Analysis

2.1 One predictor

The Gaussian density in class k is $f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\{-\frac{1}{2}(\frac{x-\mu_k}{\sigma_k})^2\}$. LDA assumes $\sigma_1 = \cdots = \sigma_K = \sigma$. Plugging into Bayes' formula,

$$p_k(x) = \frac{\pi_k \exp\{-\frac{1}{2\sigma^2}(x-\mu_k)^2\}}{\sum_{l=1}^{K} \pi_l \exp\{-\frac{1}{2\sigma^2}(x-\mu_l)^2\}}.$$

Theorem: discriminant functions

Classifying to the largest $p_k(x)$ is equivalent to classifying to the largest

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k,$$

a *linear* function of x. With $K = 2$ and $\pi_1 = \pi_2 = 0.5$ the decision boundary is $x = (\mu_1 + \mu_2)/2$.

Proof. The denominator does not depend on k. Taking logs, $\log(\pi_k f_k(x)) = \log \pi_k - \frac{(x-\mu_k)^2}{2\sigma^2} + c = \log \pi_k - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + c$, and $-x^2/(2\sigma^2) + c$ is common to all classes. For two classes with equal priors, $\delta_1(x) = \delta_2(x)$ gives $x(\mu_1 - \mu_2) = (\mu_1^2 - \mu_2^2)/2$. ■

Example: simulation

$\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, $\sigma^2 = 1$: the Bayes boundary is $x = 0$, with Bayes error $\Phi(-1.5) \approx 6.7\%$. With 20 observations drawn from each class, the LDA boundary computed from the estimates lies close to 0, so its error is only slightly above the Bayes error.

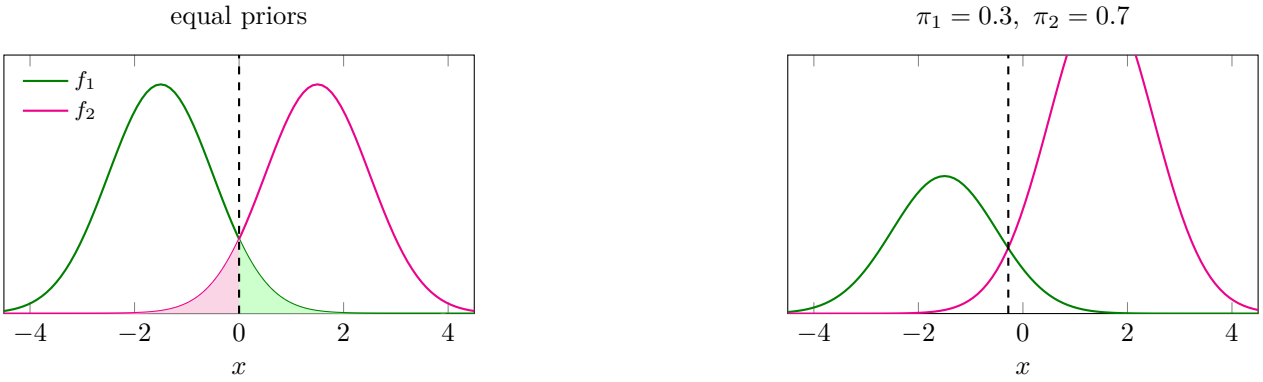


Figure 1: One-dimensional LDA with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\sigma = 1$ (curves show $\pi_k f_k$ up to a constant). Left: the Bayes boundary is $x = 0$; the shaded tails are the two misclassification probabilities. Right: favouring class 2 moves the boundary to $x = \frac{\mu_1+\mu_2}{2} + \frac{\sigma^2 \log(\pi_1/\pi_2)}{\mu_2-\mu_1} \approx -0.28$, i.e. towards class 1.

2.2 Estimating the parameters

$$\hat{\pi}_k = \frac{n_k}{n}, \qquad \hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i, \qquad \hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2 \quad \text{(pooled variance)},$$

plugged into $\hat{\delta}_k(x)$. These are the MLEs of the full (joint) likelihood, except for the unbiased divisor $n - K$.

2.3 Several predictors

The multivariate Gaussian density is

$$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\Big\{ -\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big\},$$

and LDA assumes $\Sigma_k = \Sigma$ for all k.

Theorem: LDA in p dimensions

$$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k, \qquad \hat{C}(x) = \arg\max_k \delta_k(x).$$

The boundary between classes k and k' is the hyperplane

$$\Big\{x :\ x^\top \Sigma^{-1}(\mu_k - \mu_{k'}) + \log \frac{\pi_k}{\pi_{k'}} - \frac{1}{2}(\mu_k + \mu_{k'})^\top \Sigma^{-1}(\mu_k - \mu_{k'}) = 0\Big\},$$

with normal vector $\Sigma^{-1}(\mu_k - \mu_{k'})$, generally *not* in the direction of $\mu_k - \mu_{k'}$.

Proof. As in one dimension: $\log(\pi_k f_k(x)) = \log \pi_k - \frac{1}{2}(x-\mu_k)^\top \Sigma^{-1}(x-\mu_k) + c$, and the quadratic term $-\frac{1}{2}x^\top \Sigma^{-1} x$ is shared by all classes because Σ is common. Set $\delta_k = \delta_{k'}$. ■

With $p = 2$ and $K = 3$, the three LDA boundaries (straight lines) are close to the Bayes boundaries. **Iris data**: 4 variables, 3 species, 50 samples each; LDA classifies all but 3 of the 150 training samples correctly. **Credit (Default) data**: training error 2.75%, FPR 0.2%, FNR 75.7% (Ch. 8).

2.4 Adjusting the errors

LDA classifies to Yes if $\Pr(\text{Yes} \mid \text{balance}, \text{student}) \geq \Pr(\text{No} \mid \cdot)$, i.e. with threshold 0.5. Changing the prior of Yes, or the threshold from 0.5 to another value in $[0, 1]$, trades the two error rates. Lowering the threshold to 0.2 reduces the FNR on the Default data from 75.7% to about 41% while the overall error rises only slightly; to reduce false negatives further one may go to 0.1 or less. Logistic regression is adjusted in the same way.

3 Quadratic Discriminant Analysis

QDA allows the Σ_k to differ. The discriminant function is

$$\delta_k(x) = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^\top \Sigma_k^{-1}(x-\mu_k) + \log \pi_k = x^\top W_k x + x^\top w_k + b_k,$$

with $W_k = -\frac{1}{2}\Sigma_k^{-1}$, $w_k = \Sigma_k^{-1}\mu_k$ and $b_k = -\frac{1}{2}\mu_k^\top \Sigma_k^{-1}\mu_k - \frac{1}{2}\log|\Sigma_k| + \log \pi_k$: a quadratic form. The boundary between k and k' is the quadric

$$\{x:\ x^\top (W_k - W_{k'})x + x^\top (w_k - w_{k'}) + (b_k - b_{k'}) = 0\}.$$

Parameter count. LDA estimates Kp means and one covariance, $p(p+1)/2$ parameters; QDA estimates $Kp(p+1)/2$ covariance parameters — with $p = 50$ and $K = 3$, 3825 versus 1275. QDA works best when the class covariances differ a lot and there are enough observations to estimate them; LDA works best when the covariances are similar or data are too few (lower variance, possibly higher bias). Both perform well on many real problems. With equal covariances the Bayes boundary is linear and LDA beats QDA; with different covariances QDA is closer to the Bayes boundary.

4 Naive Bayes

Assume the inputs are independent *within each class*:

$$f(x_1, \ldots, x_p \mid y = c_k) = f(x_1 \mid c_k) f(x_2 \mid c_k) \cdots f(x_p \mid c_k).$$

Gaussian naive Bayes takes each Σ_k diagonal:

$$\delta_k(x) \ \propto\ \log\Big[\pi_k \prod_{j=1}^{p} f_{kj}(x_j)\Big] = -\frac{1}{2}\sum_{j=1}^{p}\Big[\frac{(x_j - \mu_{kj})^2}{\sigma_{kj}^2} + \log \sigma_{kj}^2\Big] + \log \pi_k.$$

Estimating a p-dimensional density is hard because the dependence among predictors must be characterised; naive Bayes eliminates it, needing only one-dimensional densities (Gaussian, kernel density, or frequency tables for categorical X_j). Although the assumption is made for convenience, it often gives decent results, especially when n is not large relative to p: the bias of a wrong independence assumption is outweighed by the large reduction in variance, and classification only needs the *ranking* of posteriors to be right, not the probabilities.

5 Comparison of Classification Methods

5.1 Logistic regression versus LDA

For LDA, the log posterior odds between classes k and K are linear:

$$\log \frac{\Pr(y = c_k \mid x)}{\Pr(y = c_K \mid x)} = x^\top \Sigma^{-1}(\mu_k - \mu_K) + \log \frac{\pi_k}{\pi_K} - \frac{1}{2}(\mu_k + \mu_K)^\top \Sigma^{-1}(\mu_k - \mu_K) = \alpha_{k0} + \alpha_k^\top x,$$

and logistic regression has linear logits by construction, $\beta_{k0} + \beta_k^\top x$. The same form — are they the same estimator? No:

- **Origin of linearity.** For LDA it is a consequence of Gaussian classes with a common covariance; for logistic regression it holds by construction.
- **Common component.** The joint density factorises as $\Pr(X = x, Y = c_k) = \Pr(X = x) \Pr(Y = c_k \mid x)$, and for both models the second factor has the same multinomial-logit form $\exp(\beta_{k0} + \beta_k^\top x)/(1 + \sum_{k'} \exp(\beta_{k'0} + \beta_{k'}^\top x))$.
- **Different assumptions on $\Pr(X)$.** Logistic regression leaves the marginal of X arbitrary; LDA assumes the Gaussian mixture $\Pr(X = x) = \sum_k \pi_k \phi(x; \mu_k, \Sigma)$. Logistic regression makes fewer assumptions and is more general.
- **Estimation.** Logistic regression maximises the *conditional* likelihood of Y given X, ignoring $\Pr(x)$ (fully nonparametric: the empirical distribution with mass $1/n$ at each point). LDA maximises the *full* likelihood of $\phi(x; \mu_k, \Sigma)\pi_k$, so the marginal density does play a role.

Remarks. LDA is easier to compute. If the class densities really are Gaussian, LDA is better: logistic regression may lose about 30% efficiency asymptotically in error rate (Efron, 1975). On robustness, LDA uses all points (also those far from the boundary) to estimate Σ — more information but sensitive to outliers; logistic regression down-weights points far from the boundary (their $p_i(1 - p_i)$ weights are tiny) and is more robust.

5.2 QDA versus LDA, logistic regression and KNN

QDA is a compromise between nonparametric KNN and linear LDA/logistic regression. If the true boundary is linear, LDA and logistic regression do best; moderately nonlinear, QDA does best; more complicated, KNN (with K chosen by CV) is superior — given enough data.

Method	Boundary	Strengths / weaknesses
logistic regression	linear (or by basis expansion)	few assumptions; probabilities; unstable under separation
LDA	linear	efficient under Gaussian classes; stable with small n; outlier sensitive
QDA	quadratic	flexible covariances; many parameters
naive Bayes	depends on f_{kj}	scales to large p; independence assumption
KNN	arbitrary	nonparametric; needs much data; no interpretation

Summary

- Generative classifiers model $f_k(x)$ and π_k and classify by $\arg\max_k \pi_k f_k(x)$.

- LDA (common Σ) has linear discriminants $\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k$; QDA (class-specific Σ_k) has quadratic boundaries.
- Naive Bayes assumes conditional independence of the features, trading bias for variance.
- LDA and logistic regression share the logit form but differ in assumptions on $\Pr(X)$ and in estimation (full versus conditional likelihood).

Chapter 11. Tree-Based Methods and Ensembles

Lecture 8 (regression and classification trees, pruning, bagging, OOB, random forests, boosting, AdaBoost, GBM, XGBoost, LightGBM, CatBoost, importance, SHAP)

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1 Decision Trees

Decision trees apply to both regression and classification. They *stratify* or segment the predictor space into simple regions; since the splitting rules can be summarised in a tree, these are decision-tree methods.

Pros and cons. Trees are simple and useful for interpretation, but typically not competitive with the best supervised methods in accuracy. Bagging, random forests and boosting grow many trees and combine them into a consensus prediction, often dramatically improving accuracy — at the price of interpretability.

2 Regression Trees

2.1 Example: Hitters data

Predict the log salary of a baseball player from `Years` (in the major leagues) and `Hits` (in the previous year). The tree has two internal nodes and three terminal nodes (leaves):

$$R_1 = \{X : \texttt{Years} < 4.5\}, \quad R_2 = \{X : \texttt{Years} \geq 4.5,\ \texttt{Hits} < 117.5\},$$

$$R_3 = \{X : \texttt{Years} \geq 4.5,\ \texttt{Hits} \geq 117.5\},$$

and the number in each leaf is the mean response of the observations there. The regions are **terminal nodes**; the split points are **internal nodes**; trees are drawn upside down with leaves at the bottom.

Interpretation. Years is the most important factor: less experienced players earn less. Among inexperienced players, Hits plays little role; among players with five or more years, more Hits means higher salary. This is surely an oversimplification, but compared with a regression model it is easy to display, interpret and explain — and it contains an interaction that a linear model would need to be told about.

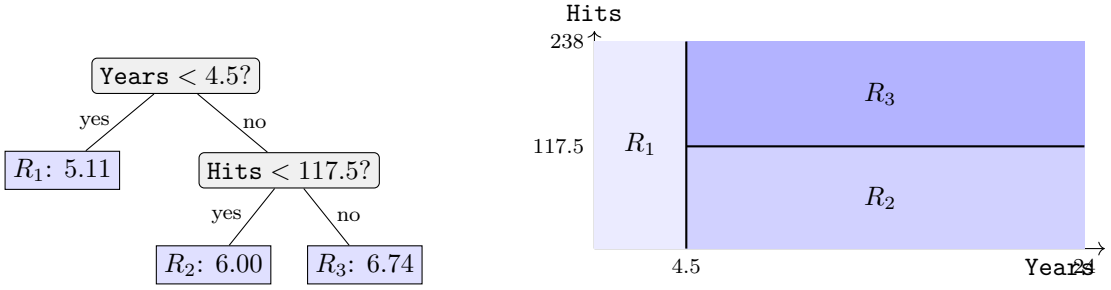


Figure 1: The Hitters regression tree (leaf values are mean log salaries) and the corresponding partition of the predictor space into three boxes.

2.2 Building a tree

1. Divide the predictor space into J distinct, non-overlapping regions $R_1, \ldots, R_J$.
2. For every observation in R_j, predict the mean response of the training observations in R_j.

In theory the regions could have any shape; for simplicity and interpretability we use high-dimensional rectangles (boxes). The goal is to find boxes minimising

$$\sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2,$$

where $\hat{y}_{R_j}$ is the mean response in box j (for fixed boxes the mean is the RSS-optimal constant).

2.3 Recursive binary splitting

Considering every partition into J boxes is computationally infeasible, so we use a **top-down, greedy** approach: top-down because it starts at the top of the tree and successively splits; greedy because at each step the best split is made *at that step*, without looking ahead.

Algorithm: recursive binary splitting

Select the predictor X_j and cutpoint s that minimise

$$\sum_{i:x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2 + \sum_{i:x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2,$$

$$R_1(j,s) = \{X : X_j < s\}, \; R_2(j,s) = \{X : X_j \geq s\}.$$

Then repeat within one of the resulting regions (not the whole space), and so on, until a stopping criterion is reached (e.g. no region contains more than five observations).

For each j the search over s is fast: sort the values and update running sums. A test observation is predicted by the mean of the training observations in its region.

2.4 Pruning

The process may predict the training set well but overfit. A smaller tree (fewer regions) may have lower variance and better interpretation at the cost of a little bias. Growing only while each split decreases RSS by more than a (high) threshold is too short-sighted: a seemingly worthless split early on may be followed by a very good split. Better: grow a very large tree T_0 and prune it back.

Definition: cost-complexity (weakest-link) pruning

For each $\alpha \geq 0$ there is a subtree $T \subset T_0$ minimising

$$\sum_{m=1}^{|T|} \sum_{i:x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|,$$

where $|T|$ is the number of terminal nodes, R_m the rectangle of the mth terminal node and $\hat{y}_{R_m}$ its mean.

α trades the subtree's complexity against its fit: $\alpha = 0$ gives T_0; as α increases there is a price for many leaves and the optimal subtree shrinks. As α increases, branches are pruned in a nested, predictable fashion, so the whole sequence of subtrees is easy to obtain: collapsing the internal node t (with branch T_t) pays off once $\alpha \geq \{R(t) - R(T_t)\}/(|T_t| - 1)$, and the weakest link (smallest such value) is pruned first.

Algorithm: building a regression tree

1. Grow a large tree by recursive binary splitting, stopping only when each terminal node has fewer than some minimum number of observations.
2. Apply cost-complexity pruning to obtain the sequence of best subtrees as a function of α.
3. Choose α by K-fold CV: for each k, repeat steps 1–2 on all but the kth fold and evaluate

the MSE on the kth fold as a function of α; average and pick $\hat{\alpha}$.

4. Return the subtree from step 2 corresponding to $\hat{\alpha}$.

Hitters: 132 training and 131 test observations; a large tree was grown and pruned, and six-fold CV estimated the MSE as a function of α. The CV error is minimised by a tree with three leaves, close to the test-error minimum.

3 Classification Trees

A classification tree predicts that each observation belongs to the most commonly occurring class of training observations in its region, and the class proportions serve as probability estimates. It is grown by recursive binary splitting, but RSS cannot be used as the criterion.

Definition: node impurity measures

With $\hat{p}_{mk}$ the proportion of training observations in region m from class k:

$$E = 1 - \max_k \hat{p}_{mk} \text{ (classification error)}, \qquad G = \sum_{k=1}^{K} \hat{p}_{mk}(1-\hat{p}_{mk}) \text{ (Gini)},$$

$$D = -\sum_{k=1}^{K} \hat{p}_{mk} \log \hat{p}_{mk} \text{ (cross-entropy)}.$$

The classification error is not sensitive enough for tree growing. The Gini index is a measure of total variance across the K classes (the sum of Bernoulli variances $\hat{p}_{mk}(1-\hat{p}_{mk})$); it is small when all $\hat{p}_{mk}$ are close to 0 or 1, hence a measure of **node purity**. Cross-entropy behaves very similarly numerically.

Example: why Gini is more sensitive than the error rate

A node with 400 cases of each class is split either into $(300, 100)$ and $(100, 300)$, or into $(200, 400)$ and $(200, 0)$. Both splits have misclassification rate 0.25. The Gini values are $\frac{1}{2}(0.375) + \frac{1}{2}(0.375) = 0.375$ and $\frac{3}{4}(0.444) + \frac{1}{4}(0) = 0.333$: Gini prefers the second split, which creates a pure node.

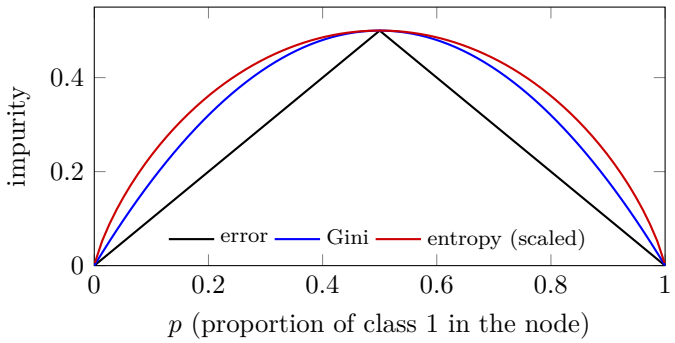


Figure 2: Node impurity for two classes. Gini $2p(1-p)$ and entropy (scaled to 0.5 at $p = \frac{1}{2}$) are strictly concave, so they reward splits that make nodes purer even when the error rate $\min(p, 1-p)$ does not change.

Because Gini and entropy are strictly concave in p, any split weakly decreases the weighted impurity (Jensen's inequality), whereas the piecewise-linear error rate often shows no improvement. For pruning, the classification error is preferred if prediction accuracy is the goal.

Heart data: binary outcome HD for 303 patients with chest pain (13 predictors: Age, Sex, Chol, ...). CV yields a tree with six terminal nodes. Trees handle qualitative predictors directly (a split assigns some levels to one branch).

3.1 Trees versus linear models

Linear regression assumes $f(X) = \beta_0 + \sum_j \beta_j X_j$; a regression tree assumes $f(X) = \sum_{m=1}^{M} c_m \mathbb{I}\{X \in R_m\}$. If the true relationship is close to linear, the linear model wins; if it is highly nonlinear and box-like, the tree wins. Example: the green class has two independent standard normal inputs, the red class is conditioned on $X_1^2 + X_2^2 \geq 4.6$; the circular boundary cannot be captured linearly, while a tree approximates it by a staircase of boxes.

3.2 Advantages and disadvantages of trees

Trees are very easy to explain (even easier than linear regression), may mirror human decision-making, can be displayed graphically and interpreted by non-experts, and handle qualitative predictors without dummy variables. But they generally lack the predictive accuracy of other methods and are non-robust: a small change in the data can cause a large change in the tree. Aggregating many trees substantially improves performance.

4 Bagging

4.1 Idea

Bootstrap aggregation (bagging) is a general-purpose procedure for reducing the variance of a learning method, particularly useful for trees. Given n independent observations each with variance σ^2, the mean has variance σ^2/n: averaging reduces variance. We do not have many training sets, so we bootstrap: generate B bootstrapped training sets, fit $\hat{f}^{*b}(x)$ on the bth, and average,

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}^{*b}(x).$$

For classification, record the class predicted by each of the B trees and take a **majority vote** (or average the class probabilities). Bagging can improve tree predictions a lot (and gives more flexible decision boundaries), but interpretation is harder. The trees are grown deep and not pruned: each has high variance and low bias, and averaging reduces the variance.

Theorem: variance of an average of identically distributed predictors

If $\hat{f}^{*1}, \ldots, \hat{f}^{*B}$ at a point x each have variance σ^2 and pairwise correlation ρ, then

$$\operatorname{Var}\Big(\frac{1}{B} \sum_{b} \hat{f}^{*b}(x)\Big) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Proof. $\operatorname{Var}(\sum_b \hat{f}^{*b}) = B\sigma^2 + B(B-1)\rho\sigma^2$; divide by B^2. ■

As B grows the second term vanishes but the first remains: bagged trees built on overlapping bootstrap samples are positively correlated, which limits the gain. Increasing B does not lead to overfitting.

4.2 Out-of-bag error estimation

Theorem: each bagged tree uses about two thirds of the observations

The probability that a given observation is *not* in a bootstrap sample is $(1-1/n)^n \to e^{-1} \approx 0.368$.

Proof. Each of the n independent draws misses it with probability $1-1/n$; and $n\log(1-1/n) \to -1$. ■

The remaining third, the **out-of-bag** (OOB) observations of a tree, can be predicted by it. Each observation i is OOB for about $B/3$ trees; averaging (or voting over) those predictions gives an OOB prediction for i, and the resulting OOB error is a valid estimate of the test error. For large B it is essentially the LOOCV error of bagging, obtained without refitting. On the Heart data, the OOB errors of bagging and random forests track their test errors as functions of B, and both beat a single tree.

5 Random Forests

Random forests improve bagged trees by a small tweak that **decorrelates** the trees. As in bagging, trees are built on bootstrapped samples; but each time a split is considered, a random sample of m predictors is chosen as split candidates from the full set of p, and the split may use only one of them. A fresh sample is taken at each split; typically $m \approx \sqrt{p}$ (classification) or $p/3$ (regression).

Why it helps: with one very strong predictor, most bagged trees use it in the top split, so they look alike and their predictions are highly correlated; averaging highly correlated quantities reduces variance little ($\rho\sigma^2$ term). Forcing each split to consider a subset gives other predictors a chance, lowering ρ. $m = p$ is bagging.

Gene-expression data: 4,718 genes measured on 349 patients, 15 classes (normal or one of 14 cancer types). Using the 500 genes with largest training variance, random forests with $m < p$ improve slightly over bagging ($m = p$); a single tree has error 45.7%. As with bagging, random forests do not overfit as B increases, so a sufficiently large B is recommended.

6 Boosting

6.1 Boosting regression trees

Like bagging, boosting is a general approach. Bagging fits a separate tree to each bootstrap copy, independently of the other trees. Boosting grows the trees **sequentially**: each tree uses information from previously grown trees, and no bootstrap is used.

Algorithm: boosting for regression trees

1. Set $\hat{f}(x) = 0$ and $r_i = y_i$ for all i.
2. For $b = 1, \ldots, B$: fit a tree $\hat{f}^b$ with d splits ($d+1$ leaves) to the data (X, r); update $\hat{f}(x) \leftarrow \hat{f}(x) + \lambda\hat{f}^b(x)$ and the residuals $r_i \leftarrow r_i - \lambda\hat{f}^b(x_i)$.
3. Output $\hat{f}(x) = \sum_{b=1}^{B} \lambda\hat{f}^b(x)$.

Instead of fitting one large tree (fitting the data hard, potentially overfitting), boosting **learns slowly**: small trees fitted to the residuals improve $\hat{f}$ gradually in areas where it does poorly,

and the shrinkage λ slows the process further, letting more and different-shaped trees attack the residuals.

Three tuning parameters.

- The number of trees B: unlike bagging and random forests, boosting can overfit if B is too large (slowly); choose B by CV.
- The shrinkage λ: the learning rate, typically 0.01 or 0.001; very small λ needs very large B.
- The number of splits d: the interaction depth. Often $d = 1$ works well (stumps, giving an additive model); d splits allow interactions of up to d variables.

On the gene-expression data, boosted stumps ($\lambda = 0.01$) slightly outperform depth-2 boosting and random forests.

6.2 AdaBoost

Algorithm: AdaBoost, $y \in \{-1, +1\}$

Start with weights $D_1(i) = 1/n$. For $b = 1, \dots, B$: (i) train a weak learner h_b using weights D_b; (ii) compute its weighted error $\epsilon_b = \sum_i D_b(i)\mathbb{I}\{h_b(x_i) \neq y_i\}$; (iii) set $\alpha_b = \frac{1}{2}\log\frac{1-\epsilon_b}{\epsilon_b}$; (iv) update $D_{b+1}(i) = D_b(i)\exp\{-\alpha_b y_i h_b(x_i)\}/Z_b$, Z_b a normaliser. Output $H(x) = \text{sign}\big(\sum_{b=1}^{B}\alpha_b h_b(x)\big)$.

Misclassified observations get larger weights in the next round, so later learners concentrate on the hard cases; accurate learners get a larger vote.

Theorem: AdaBoost minimises the exponential loss stagewise

With $F_{b-1} = \sum_{s<b}\alpha_s h_s$, the pair (α_b, h_b) minimises $\sum_i \exp\{-y_i(F_{b-1}(x_i) + \alpha h(x_i))\}$.

Proof. With $D_b(i) \propto e^{-y_i F_{b-1}(x_i)}$, the objective is proportional to $e^{-\alpha}(1-\epsilon) + e^{\alpha}\epsilon$, which for $\alpha > 0$ is increasing in the weighted error ϵ of h (so h_b minimises it), and setting the α-derivative to zero gives $e^{2\alpha} = (1-\epsilon)/\epsilon$. The new weights are $e^{-y_i F_b(x_i)}$, i.e. the update rule. ■

In the classic toy example the three stumps have $\epsilon = 0.30, 0.21, 0.14$ and votes $0.42, 0.65, 0.92$; their weighted combination gives a boundary no single stump can draw.

6.3 Gradient boosting machines

GBM views boosting as gradient descent in function space with an arbitrary differentiable loss L: (i) at step b compute pseudo-residuals $r_{ib} = -\partial L(y_i, F(x_i))/\partial F(x_i)$ at $F = F_{b-1}$; (ii) fit a tree to them; (iii) update $F_b = F_{b-1} + \lambda h_b$. For squared loss $\frac{1}{2}(y-F)^2$ the pseudo-residual is the ordinary residual $y - F$, recovering the algorithm above (the slide writes $y = f_b(x) + R_b$, $R_b = y - f_b(x)$, and the next tree targets R_b).

Problem	Loss	Negative gradient
regression	squared $\frac{1}{2}(y-F)^2$; absolute $\lvert y-F\rvert$; Huber	$y - F$; $\text{sign}(y-F)$; clipped residual
classification	Bernoulli deviance $\log(1+e^{-2yF})$; exponential e^{-yF}	$2y/(1+e^{2yF})$; ye^{-yF}

Standard GBM is powerful but can overfit with too many trees, has no explicit regularisation of tree complexity, and sequential fitting can be computationally inefficient.

6.4 XGBoost, LightGBM, CatBoost

XGBoost is an optimised, regularised gradient boosting with objective

$$\mathcal{L} = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{b=1}^{B} \Omega(f_b), \qquad \Omega(f_b) = \gamma T_b + \frac{\lambda}{2} \sum_{j=1}^{T_b} w_{bj}^2,$$

where T_b is the number of leaves of tree f_b and w_{bj} the score of leaf j. A second-order Taylor expansion with gradients g_i and Hessians h_i gives the optimal leaf weight $w_j^* = -\sum_{i \in I_j} g_i / (\sum_{i \in I_j} h_i + \lambda)$ and a closed-form split gain. It improves regularisation (penalising complex trees), optimisation (gradients and Hessians) and systems (parallelisation, efficient pruning, cache-aware computation).

LightGBM targets scalability: histogram-based learning bins continuous features to speed up split finding; leaf-wise growth expands the leaf with the largest loss reduction (reducing loss more aggressively); it cuts memory and training time on large, high-dimensional data.

CatBoost targets categorical data: native categorical handling via target statistics computed in a principled (ordered) way, and *ordered boosting*, which prevents target leakage and prediction shift; it is robust and less prone to overfitting on small data.

Method	Main focus	Mechanism
XGBoost	regularisation and second-order optimisation	controls tree complexity; uses gradients and Hessians
LightGBM	speed and scalability	histogram-based splitting, leaf-wise growth
CatBoost	categorical data, bias reduction	ordered target statistics and ordered boosting

All three are improved versions of GBM.

7 Variable Importance

7.1 Impurity-based importance

Each split reduces a loss (RSS, Gini); a variable is important if it appears often in splits and produces large improvements. For tree-based models

$$\text{Imp}(X_j) = \sum_{\text{splits on } X_j} \text{reduction in loss}, \qquad \text{final importance} = \frac{1}{B} \sum_{b=1}^{B} \text{Imp}_b(X_j),$$

using RSS for regression and Gini or cross-entropy for classification. On the Heart data, the largest mean decreases in Gini belong to Thal, Ca and ChestPain. Importance gives a global view of the model (variable selection, interpretation, feature engineering) but does not capture causal relationships.

Limitations: biased toward variables with many possible splits (continuous or high-cardinality); depends on the model structure and configuration; not comparable across models. We need model-agnostic, more reliable measures.

7.2 Permutation importance

Break the relationship between X_j and Y by randomly shuffling X_j and measure the increase in prediction error:

$$\text{Imp}(X_j) = \text{Error}_{\text{perm}(j)} - \text{Error}_{\text{original}}.$$

It is model-agnostic and tied directly to predictive performance (compute it on OOB or validation data; with correlated predictors, a substitute variable can mask the importance of the shuffled one).

7.3 SHAP values

SHAP (Shapley additive explanations) distributes a prediction fairly among the features:

$$f(x) = \phi_0 + \sum_{j=1}^{p} \phi_j, \qquad \phi_j = \sum_{S \subseteq \{1,\dots,p\} \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \big[v(S \cup \{j\}) - v(S)\big],$$

where $v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$ is the prediction using only the features in S and $\phi_0 = \mathbb{E}f(X)$. ϕ_j is the average marginal contribution of X_j over all orders in which features can be added. Shapley values are the unique attribution satisfying efficiency ($\sum_j \phi_j = f(x) - \phi_0$), symmetry, dummy and additivity, which makes them consistent and theoretically justified; they give *local* explanations per observation (TreeSHAP computes them efficiently for trees).

Method	Type	Characteristic
tree importance	model-specific	fast but biased
permutation	model-agnostic	based on prediction error
SHAP	model-agnostic	local and consistent explanations

Summary

- Trees split the predictor space by recursive binary splitting (RSS for regression, Gini/entropy for classification) and are pruned by cost complexity with CV.
- Bagging averages deep bootstrap trees; OOB observations (about one third) estimate test error; random forests decorrelate trees by sampling m predictors at each split.
- Boosting fits small trees sequentially to residuals with shrinkage; B, λ, d are tuning parameters; AdaBoost minimises exponential loss, GBM follows the negative gradient; XGBoost, LightGBM, CatBoost refine GBM.
- Importance: impurity-based (fast, biased), permutation (model-agnostic), SHAP (local, axiomatic).

Chapter 12. Support Vector Machines

Lecture 9 (hyperplanes, maximal margin classifier, support vector classifier, kernels, multiclass SVM, SVM vs logistic regression)

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1 Separating Hyperplanes

1.1 Motivation

In a two-class problem with predictors X_1, X_2, suppose the classes are *linearly separable*: a straight line puts all points of one class on one side and all of the other class on the other. A natural approach is to find the line that gives the *biggest separation*, the points being as far from it as possible. This is the basic idea of support vector classifiers. For the general two-class problem we try to find a plane that separates the classes in feature space; if we cannot, we get creative in two ways: we soften what "separates" means, and we enrich and enlarge the feature space so that separation becomes possible.

1.2 Hyperplanes

A hyperplane in p dimensions is a flat affine subspace of dimension $p-1$:

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0;$$

in $p = 2$ dimensions it is a line. With $f(X) = \beta_0 + \beta^\top X$, $f(X) > 0$ on one side and $f(X) < 0$ on the other. Coding $y_i = +1$ for one class and $y_i = -1$ for the other, $f(X) = 0$ is a **separating hyperplane** if $y_i f(x_i) > 0$ for all i; then classify x by $\operatorname{sign} f(x)$, and $|f(x)|$ measures confidence.

Theorem: distance to a hyperplane

The signed distance of x to $\{f = 0\}$ is $f(x)/\|\beta\|$; if $\|\beta\| = 1$, $f(x)$ itself is the signed distance.

Proof. β is normal to the hyperplane. For a point x_0 on it, the distance is the length of the projection of $x - x_0$ on $\beta/\|\beta\|$: $\beta^\top(x - x_0)/\|\beta\| = (f(x) - f(x_0))/\|\beta\| = f(x)/\|\beta\|$. ■

2 The Maximal Margin Classifier

If the data are separable there are infinitely many separating hyperplanes. The **margin** M is the minimum perpendicular distance from the observations to the hyperplane; the **optimal separating hyperplane** maximises it:

$$\max_{\beta_0,\beta} M \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1, \qquad y_i(\beta_0 + \beta^\top x_i) \geq M \quad \forall i.$$

Theorem: equivalent convex problem

The problem is equivalent to

$$\min_{\beta_0,\beta} \frac{1}{2}\|\beta\|^2 \quad \text{subject to} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1 \quad \forall i,$$

with margin $M = 1/\|\beta\|$.

Proof. Drop the norm constraint and write the condition as $y_i(\beta_0 + \beta^\top x_i)/\|\beta\| \geq M$ (distances, by the previous theorem). The scale of (β_0, β) is arbitrary, so fix it by $\|\beta\| = 1/M$; then the constraints become $y_i(\beta_0 + \beta^\top x_i) \geq 1$ and maximising $M = 1/\|\beta\|$ is minimising $\|\beta\|^2/2$, a convex quadratic program. ■

The solution depends only on the points lying on the margin, the **support vectors**: moving any other point (without crossing the margin) does not change the hyperplane.

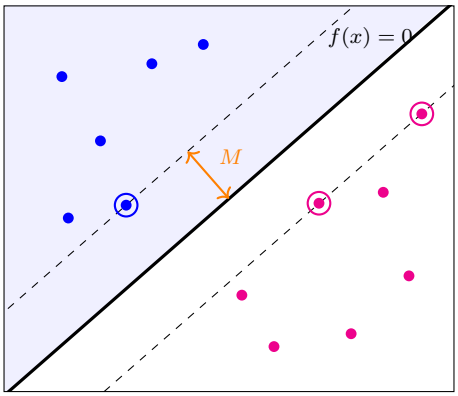


Figure 1: Maximal margin hyperplane (solid) and margins (dashed) for separable data. The circled points lie on the margins: they are the support vectors, and only they determine the hyperplane. M is the distance from the hyperplane to the margin (schematic).

2.1 Non-separable and noisy data

Often the data are not separable by a linear boundary (usually the case unless $n < p$), and then the problem has no solution. Even when separable, data may be noisy: a single observation can move the maximal margin hyperplane dramatically and produce a tiny margin. We want greater robustness to individual observations and better classification of *most* training observations: the support vector classifier maximises a **soft margin**.

3 The Support Vector Classifier

Definition: soft-margin optimisation

$$\max_{\beta_0,\beta,\epsilon} M \quad \text{subject to} \quad \sum_j \beta_j^2 = 1, \quad y_i(\beta_0 + \beta^\top x_i) \geq M(1-\epsilon_i), \quad \epsilon_i \geq 0, \quad \sum_{i=1}^n \epsilon_i \leq C.$$

ϵ_i are **slack variables**; $C \geq 0$ is a tuning parameter.

The slacks allow individual observations to violate the margin: $\epsilon_i = 0$ means observation i is on the correct side of the margin; $0 < \epsilon_i \leq 1$, on the wrong side of the margin but the correct side of the hyperplane; $\epsilon_i > 1$, on the wrong side of the hyperplane (misclassified). Since $\sum \epsilon_i \leq C$, at most C training observations can be misclassified.

C is a **budget** for violations. As C increases we tolerate more violations and the margin widens; as C decreases the margin narrows; $C = 0$ gives the maximal margin classifier. C is chosen by CV and controls the bias–variance trade-off: large C means many support vectors, low variance and possibly high bias; small C the opposite. Only observations on or inside the margin (or misclassified) are support vectors and affect the solution, so the classifier is robust to points far from the boundary — unlike LDA, which uses all points.

Derivation: the equivalent penalised form and the dual

In the parametrisation of the previous section, the soft-margin problem is

$$\min_{\beta_0,\beta,\xi} \frac{1}{2}\|\beta\|^2 + \gamma \sum_i \xi_i \quad \text{s.t.} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1 - \xi_i, \ \xi_i \geq 0,$$

where a *large* cost γ corresponds to a *small* budget C (software usually calls γ "C" — the

opposite role). The Lagrangian dual is

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2}\sum_{i,i'} \alpha_i \alpha_{i'} y_i y_{i'} \langle x_i, x_{i'} \rangle \quad \text{s.t.} \quad 0 \le \alpha_i \le \gamma, \quad \sum_i \alpha_i y_i = 0,$$

and $\hat{\beta} = \sum_i \hat{\alpha}_i y_i x_i$. By complementary slackness $\hat{\alpha}_i > 0$ only for points with $y_i f(x_i) \le 1$, the support vectors. Hence $f(x) = \hat{\beta}_0 + \sum_{i \in \mathcal{S}} \hat{\alpha}_i y_i \langle x, x_i \rangle$, and the data enter only through inner products.

4 Support Vector Machines

4.1 Feature expansion

Sometimes a linear boundary won't work, whatever C. As in regression (interactions, polynomials, splines), enlarge the feature space with transformations, e.g. use $(X_1, X_2, X_1^2, X_2^2, X_1X_2)$ instead of (X_1, X_2). A linear boundary in the enlarged space,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 = 0,$$

is nonlinear (a conic) in the original space. With cubic polynomials the boundary is $\beta_0 + \cdots + \beta_6 X_1^3 + \beta_7 X_2^3 + \beta_8 X_1 X_2^2 + \beta_9 X_1^2 X_2 = 0$. Polynomials, especially high-dimensional ones, get wild rather fast; kernels introduce nonlinearities in a more elegant and controlled way.

4.2 Inner products and kernels

The inner product is $\langle x_i, x_{i'} \rangle = \sum_{j=1}^p x_{ij} x_{i'j}$. The linear support vector classifier can be written

$$f(x) = \beta_0 + \sum_{i=1}^n \alpha_i \langle x, x_i \rangle \qquad (n+1 \text{ parameters}),$$

and estimating (β_0, α) requires only the $\binom{n}{2}$ inner products between pairs of observations. Most $\hat{\alpha}_i$ are zero:

$$f(x) = \hat{\beta}_0 + \sum_{i \in \mathcal{S}} \hat{\alpha}_i \langle x, x_i \rangle,$$

where $\mathcal{S}$ is the support set, $\{i : \hat{\alpha}_i \neq 0\}$. If we can compute inner products between observations, we can fit an SV classifier; replacing $\langle x, x_i \rangle$ by a **kernel** $K(x, x_i)$ — an inner product $\langle \Phi(x), \Phi(x_i) \rangle$ in some feature space — gives the **support vector machine**

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \hat{\alpha}_i K(x, x_i).$$

- **Polynomial kernel** of degree d: $K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^p x_{ij} x_{i'j}\right)^d$, which computes the inner products needed for d-dimensional polynomials.
- **Radial kernel**: $K(x_i, x_{i'}) = \exp\left(-\gamma \sum_{j=1}^p (x_{ij} - x_{i'j})^2\right)$, $\gamma > 0$: an implicit, infinite-dimensional feature space. Training points far from x contribute almost nothing, so the classifier is very local; γ controls the variance by squashing most dimensions severely (large γ, more flexible).

Example: the polynomial kernel is an inner product

For $p = 2$, $d = 2$: $(1+x^\top z)^2 = 1+2x_1z_1+2x_2z_2+x_1^2z_1^2+x_2^2z_2^2+2x_1x_2z_1z_2 = \langle \Phi(x), \Phi(z)\rangle$ with $\Phi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2)$: the six features of a quadratic expansion, computed without forming them.

Kernel versus enlarged feature space. Computationally we only need $K(x_i, x_{i'})$ for the $\binom{n}{2}$ distinct pairs, whereas an explicit enlarged space can be intractable (the number of degree-d monomials grows like p^d), and some kernels (the radial kernel) correspond to no finite set of features. A function K is a valid kernel iff every Gram matrix $(K(x_i, x_{i'}))$ is positive semidefinite (Mercer).

Heart data. On the training set, ROC curves show the SV classifier slightly better than LDA, and radial SVMs with larger γ nearly perfect — overfitting; on the 90 test observations the radial SVM with large γ is worst, while the SV classifier, LDA and the radial SVM with small γ perform similarly.

5 SVMs With More Than Two Classes

The SVM as defined works for $K = 2$. Two popular extensions:

- **One versus all (OVA)**: fit K two-class SVMs $\hat{f}_k(x)$, each class versus the rest; classify x to the class with the largest $\hat{f}_k(x)$.
- **One versus one (OVO)**: fit all $\binom{K}{2}$ pairwise classifiers $\hat{f}_{kl}(x)$; classify x to the class that wins the most pairwise competitions.

If K is not too large, use OVO (each fit uses only two classes and is balanced).

6 SVM versus Logistic Regression

Theorem: SVM as loss plus penalty

The support vector classifier with $f(X) = \beta_0 + \beta_1X_1 + \cdots + \beta_pX_p$ solves

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} \underbrace{\max[0,\, 1 - y_i f(x_i)]}_{\text{hinge loss}} + \lambda \sum_{j=1}^{p} \beta_j^2,$$

with large λ corresponding to a large budget C.

Proof. In the penalised form, at the optimum each slack equals $\xi_i = \max[0, 1-y_if(x_i)]$ (the smallest feasible value). Substituting gives $\frac{1}{2}\|\beta\|^2 + \gamma \sum_i \max[0, 1 - y_if(x_i)]$; divide by γ and set $\lambda = 1/(2\gamma)$. ■

This is the "loss + penalty" form, like ridge regression. The hinge loss is exactly zero for observations on the correct side of the margin ($y_if \geq 1$), which is why only support vectors matter. It is very similar to the logistic-regression loss (negative log-likelihood) $\log(1+e^{-y_if(x_i)})$, which is never exactly zero but small for $y_if \gg 0$; hence the two often give similar results.

Which to use?

- When the classes are (nearly) separable, SVM does better than logistic regression; so does LDA.

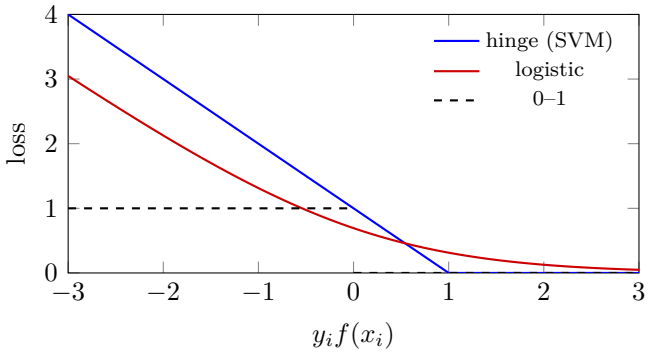


Figure 2: Hinge loss $\max(0, 1 - yf)$ is exactly zero once a point is beyond the margin ($yf \geq 1$); the logistic loss $\log(1 + e^{-yf})$ is a smooth version that never reaches zero. Both are convex surrogates of the 0–1 loss.

- When not, logistic regression (with a ridge penalty) and SVM are very similar.
- To estimate probabilities, logistic regression is the choice.
- For nonlinear boundaries, kernel SVMs are popular; kernels can be used with logistic regression and LDA as well, but computations are more expensive.

Summary

- A separating hyperplane classifies by $\text{sign} f(x)$; the maximal margin classifier maximises $M = 1/\|\beta\|$ and depends only on support vectors.
- The support vector classifier allows margin violations with slacks under a budget C (chosen by CV).
- The solution depends on the data only through inner products, so kernels (polynomial, radial) give nonlinear SVMs without explicit feature expansion.
- OVA and OVO extend SVMs to $K > 2$ classes.
- SVM = hinge loss + ridge penalty, close to penalised logistic regression; use logistic regression when probabilities are needed.